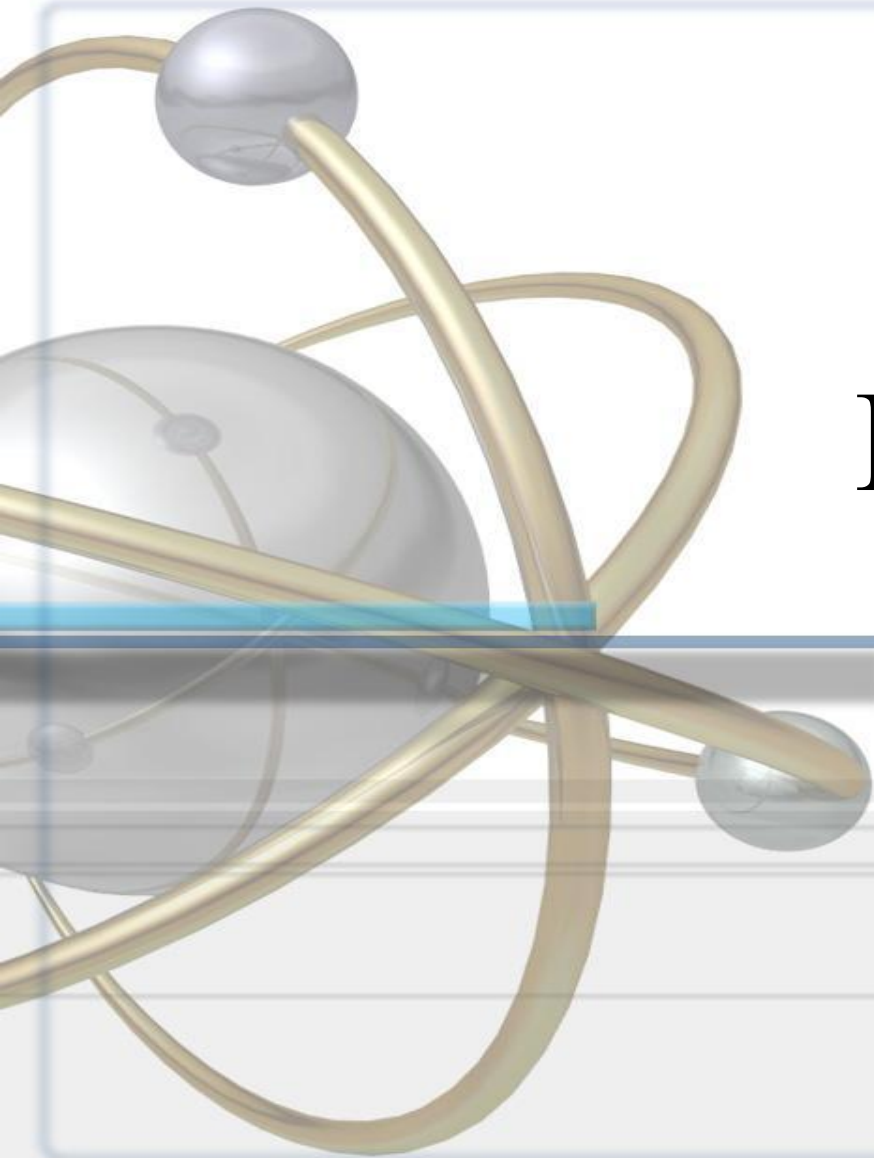


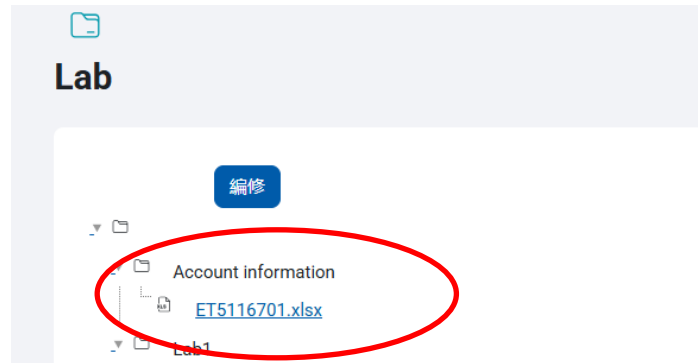


Laboratory 1

CMOS Inverter

Lab 1: Current Reference

Please refer to this file (ET5116701.xlsx) for account information



Please change your password after logging in

```
[et5115701_12@EDA Server-A ~]$ passwd
Changing password for user et5115701_12.
Current password:
New password:
Retype new password:
passwd: all authentication tokens updated successfully.
[et5115701_12@EDA Server-A ~]$
```





Lab 1: Current Reference

All content on the server, including lecture notes, process files, and other related documents, is for viewing only on the server.

Please do not download to your personal computer without permission, and do not print or take photos.

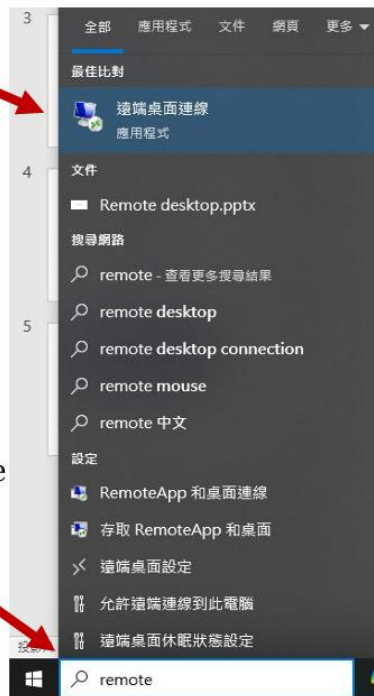
Please exercise caution and comply with relevant regulations.
Please bear the relevant legal responsibilities yourself!!!



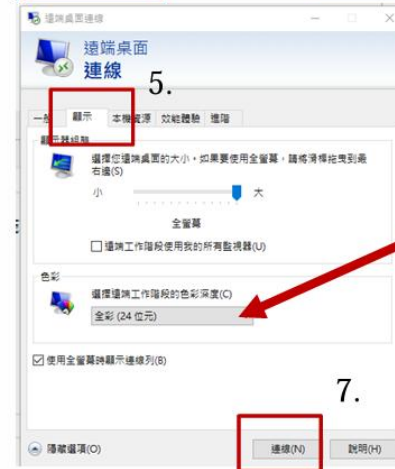
Login

2. Choose this

1. Enter remote



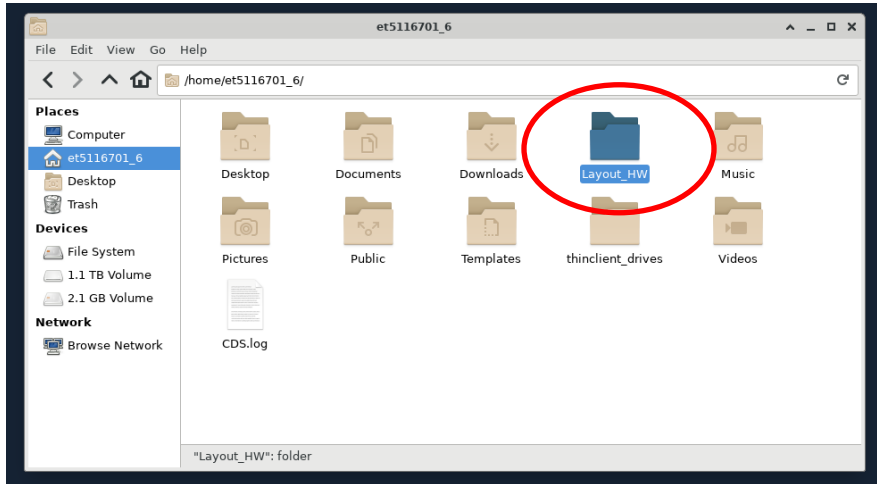
3. Enter IP



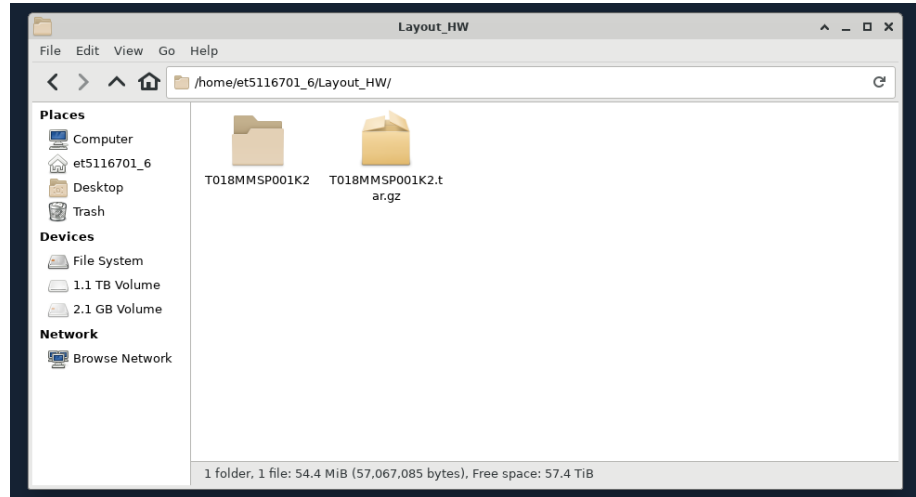
6. Select 24bit

7.

Technology Library



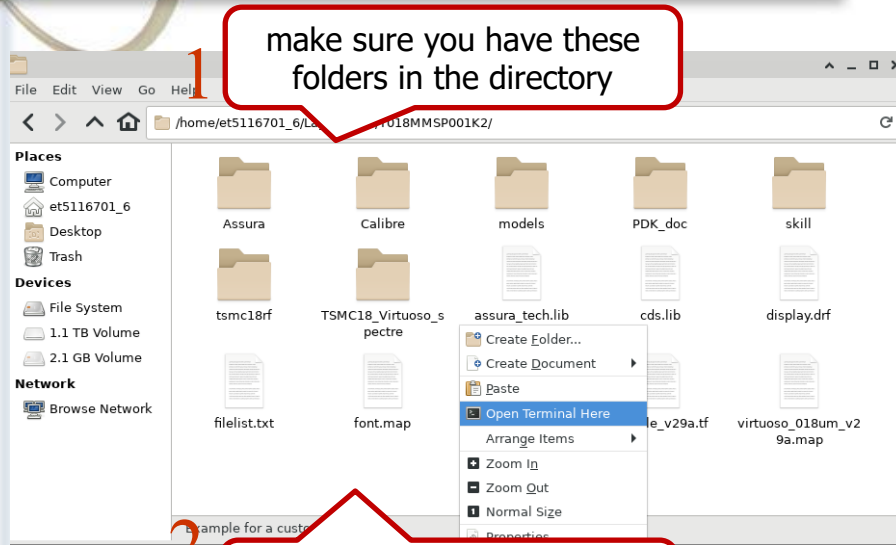
1. Create Layout_HW folder in your account



2. Copy file "**T018MMSP001K2.tar.gz**" from /File System/tools/cad/T018MMSP001K2.tar.gz to Layout_HW folder and extract it



Open Virtuoso (1/5)



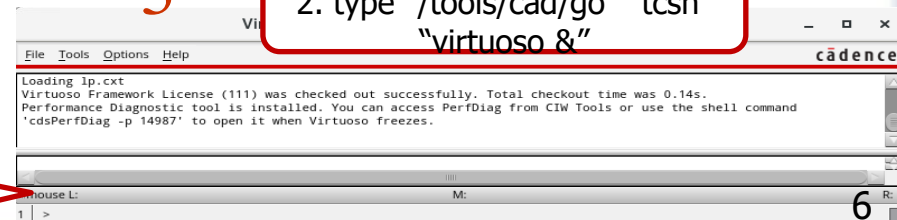
```
Terminal - et5116701_6@eda_system:T018MMSPO01K2
File Edit View Terminal Tabs Help
[et5116701_6@eda_system T018MMSPO01K2] /tools/cad/go
[et5116701_6@eda_system T018MMSPO01K2]$ tcsh
INFO:: set EDA Tools(version)
set ICADVM version: ICADVM 20.10.340 (default)
set SPECTRE version: SPECTRE 23.10.477 (default)
/bin/ls: cannot access '/usr/cad/cadence/INNOVUS': No such file or directory
basename: missing operand
Try 'basename --help' for more information.
set INNOVUS version: (default)
set DDI version: DDIEXPORT 23.32.000 (default)
set GENUS version: GENUS 21.18.000 (default)
set synthesis version: 2024.09-sp2 (default)
set verdi version: 2024.09-sp2 (default)
set vcs version: 2024.09-sp1/ (default)
set lc version: 2024.09-sp2 (default)
set fc version: 2024.09-sp2/ (default)
set icc2 version: 2024.09-sp2/ (default)
set customcompiler version: 2024.09-sp1/ (default)
set hspice version: 2024.09-sp1/ (default)
set customexplorer version: 2024.09-sp1/ (default)
set icvalidator version: 2024.09-sp2/ (default)
set icv_wb version: 2024.09-sp1/ (default)
set star-rcxt version: 2024.09-sp2/ (default)
set calibre version: aok cal 2022.1 36.16 (default)
[et5116701_6@eda_system T018MMSPO01K2]$ virtuoso &
[1] 12003
[et5116701_6@eda_system T018MMSPO01K2]$
```

3

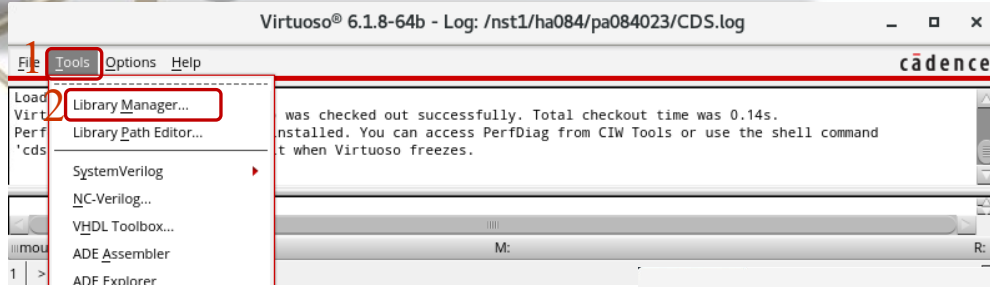
1. the terminal will pop up
2. type "/tools/cad/go" "tcsh" "virtuoso &"

4

the CIW (command interpreter window) will pop up

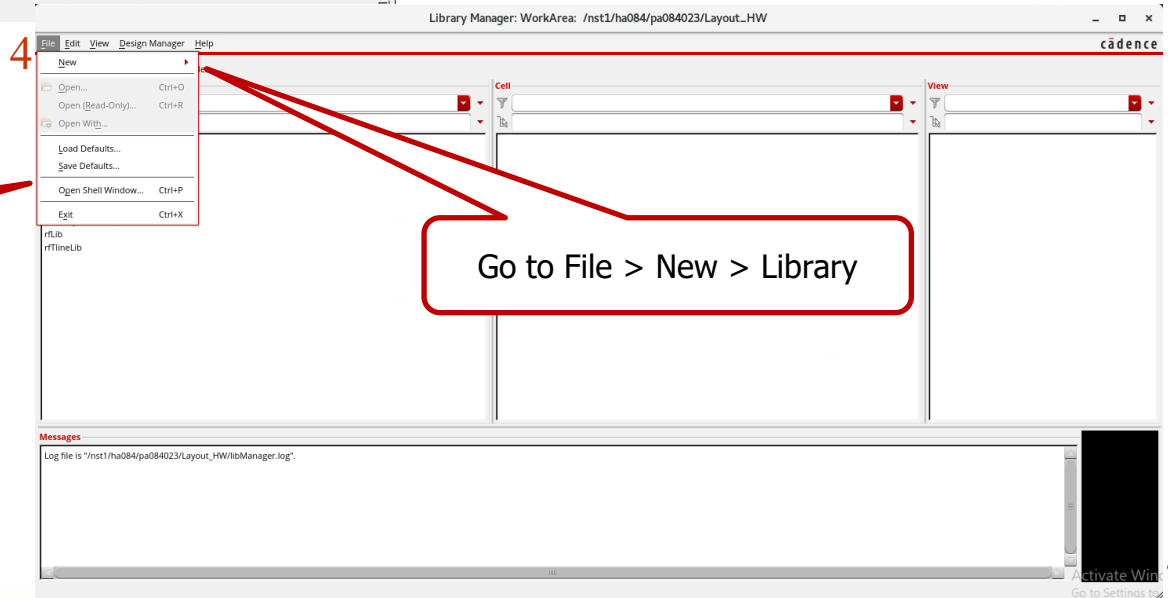


Open Virtuoso (2/5)



Tools > Library Manager

3 Library Manager will pop up

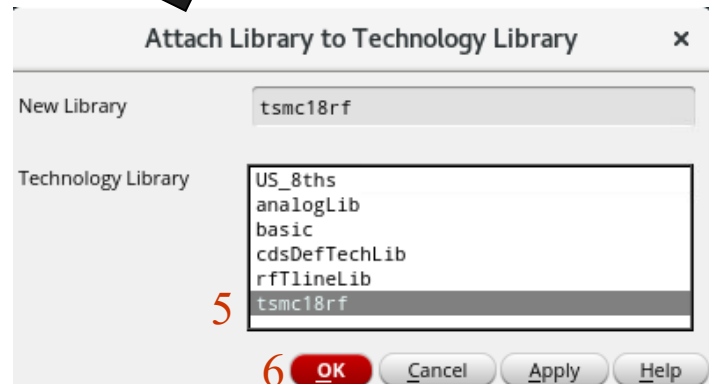
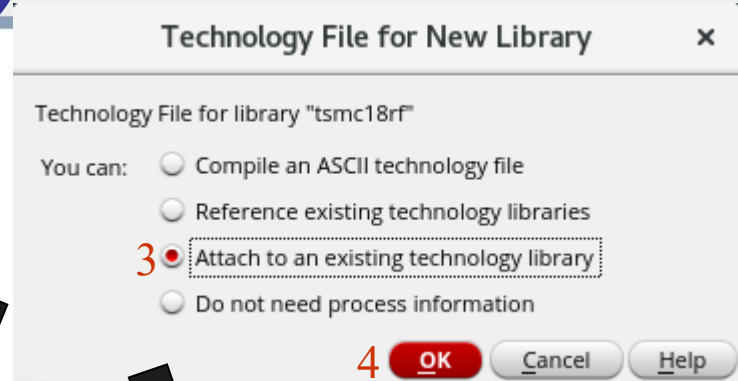
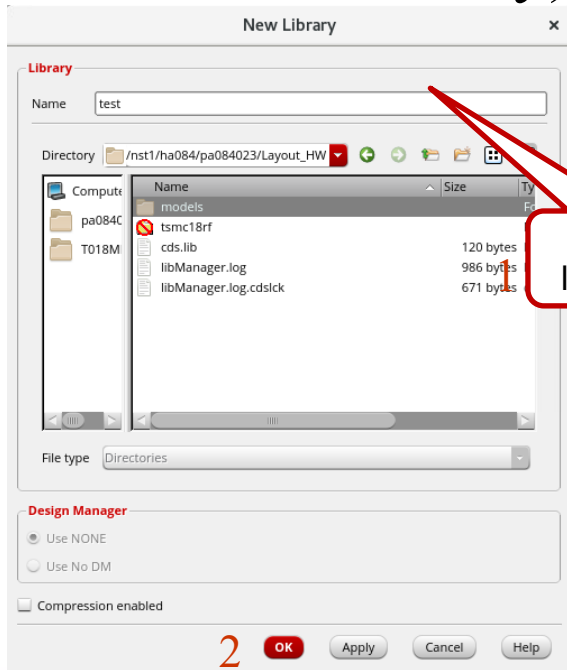


Go to File > New > Library



Open Virtuoso (4/5)

create a new library for your project
(File > New > Library)





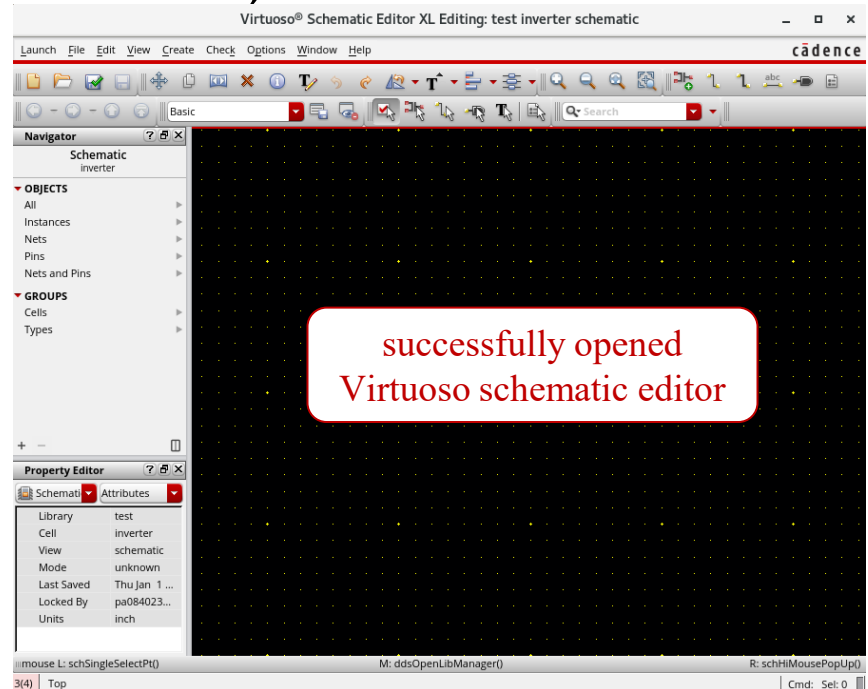
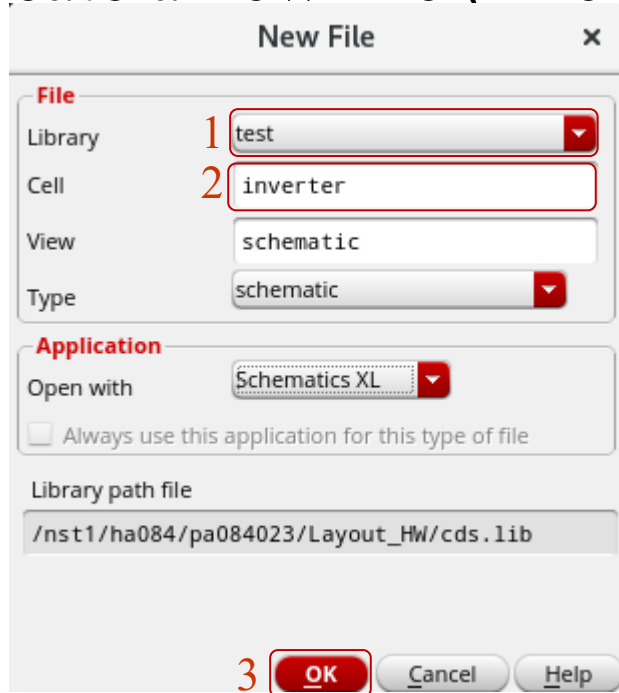
Virtuoso Common Functions

- Properties hotkey “q”
- Path hotkey “p” : Draw a fixed-width layout
- Rectangle hotkey “r” : Draw rectangle layout
- Ruler hotkey “k” : Measure the distance between two points ,
To clean it with “shift + k”
- Stretch hotkey “s” : change length and width
- Undo hotkey “u”
- Copy hotkey “c”
- Move hotkey “m”
- Align hotkey “a”



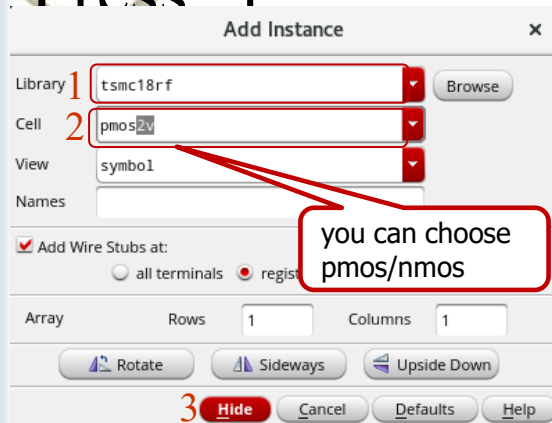
Open Virtuoso (5/5)

create a new file (File > New > Cell)



Create Transistor

Press “i”



Library 1 tsmc18rf

Cell 2 pmos2

View symbol

Names

☒ Add Wire Stubs at:
☐ all terminals ☒ registers

Array Rows 1 Columns 1

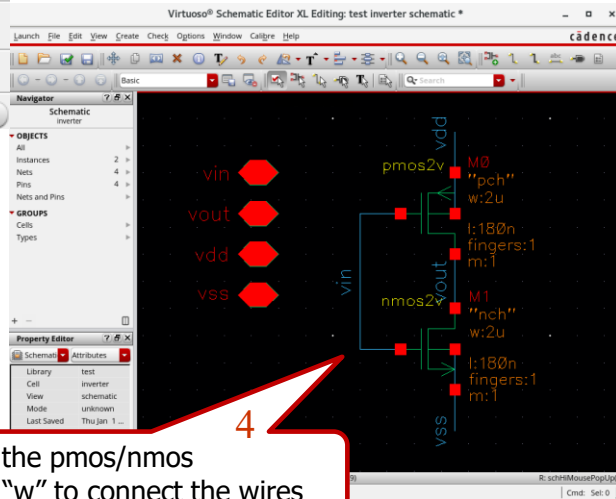
3 Hide Cancel Defaults Help

you can choose pmos/nmos

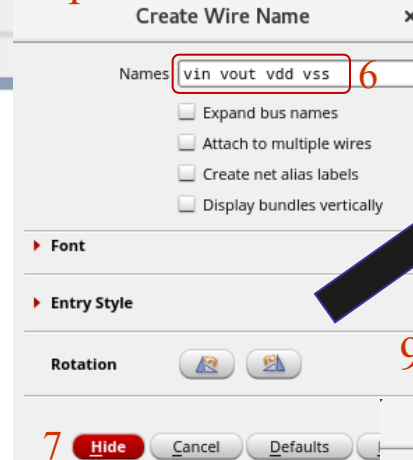
you can press “q” to edit the parameters of the devices



1. place the pmos/nmos
2. press “w” to connect the wires



5 press “l” to create labels



Create Wire Name

Names vin vout vdd vss 6

☐ Expand bus names

☐ Attach to multiple wires

☐ Create net alias labels

☐ Display bundles vertically

Font

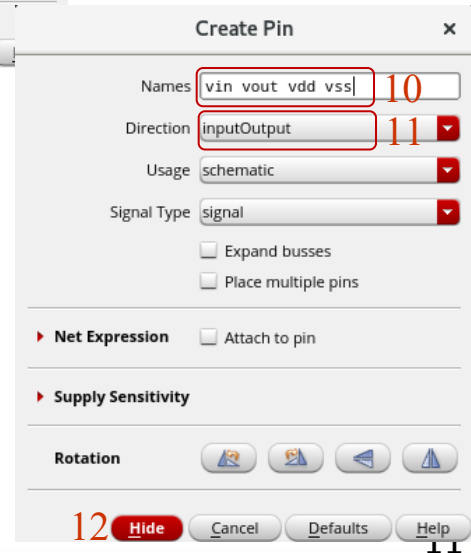
Entry Style

Rotation

7 Hide Cancel Defaults

8 attach the labels to the net

9 press “p” to create pins



Create Pin

Names vin vout vdd vss 10

Direction InputOutput 11

Usage schematic

Signal Type signal

☐ Expand busses

☐ Place multiple pins

Net Expression ☐ Attach to pin

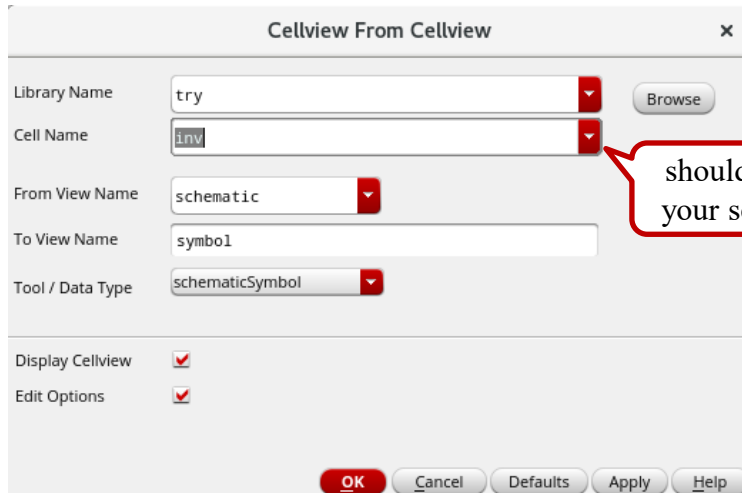
Supply Sensitivity

Rotation

12 Hide Cancel Defaults Help

Create Symbol (1/2)

Create > Cellview > From Cellview



Cellview From Cellview

Library Name: try

Cell Name: inv

From View Name: schematic

To View Name: symbol

Tool / Data Type: schematicSymbol

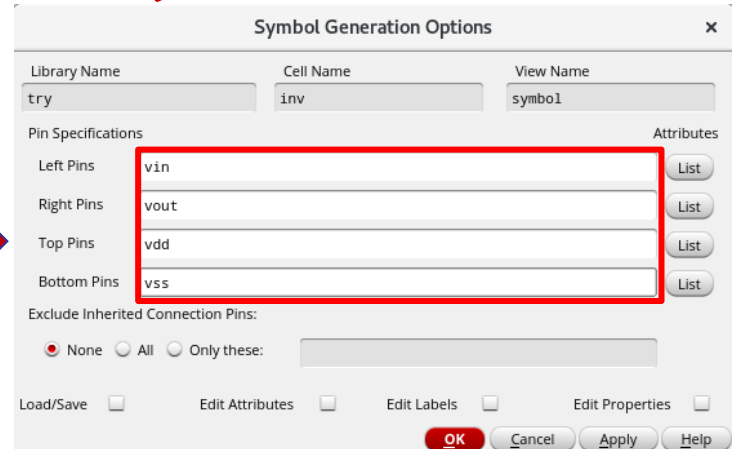
Display Cellview: ☒

Edit Options: ☒

Buttons: OK, Cancel, Defaults, Apply, Help

should be the same as your schematic design

Click "Ok"



Symbol Generation Options

Library Name: try

Cell Name: inv

View Name: symbol

Pin Specifications

	Attributes
Left Pins	vin
Right Pins	vout
Top Pins	vdd
Bottom Pins	vss

Exclude Inherited Connection Pins:

☒ None ☐ All ☐ Only these:

Buttons: Load/Save, Edit Attributes, Edit Labels, Edit Properties, OK, Cancel, Apply, Help

specify the position of the pins in the symbol

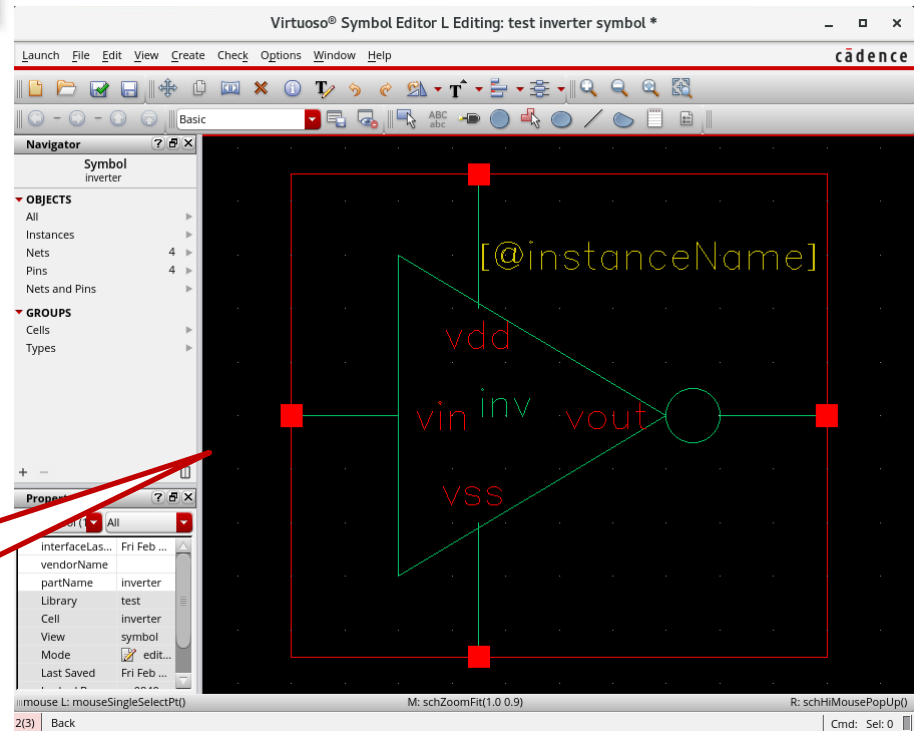
Click "Ok"



Create Symbol (2/2)

This window will pop up

you can edit the symbol as you like:
draw shapes, edit the names, etc

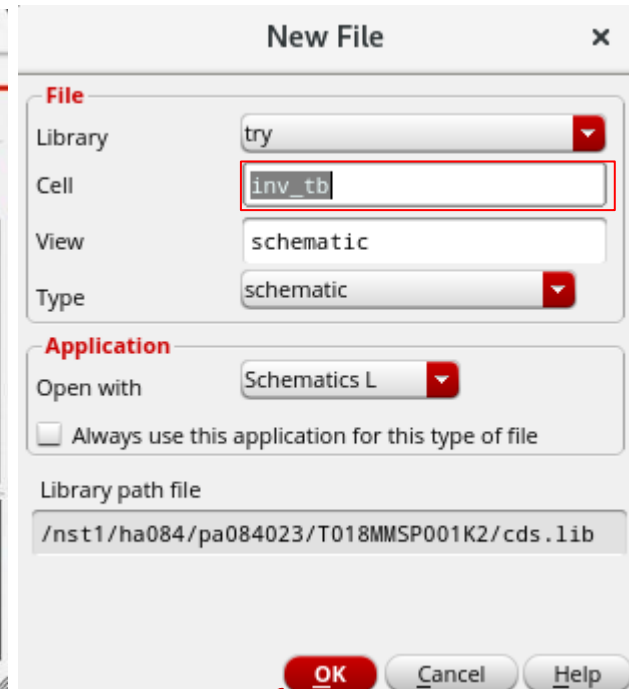
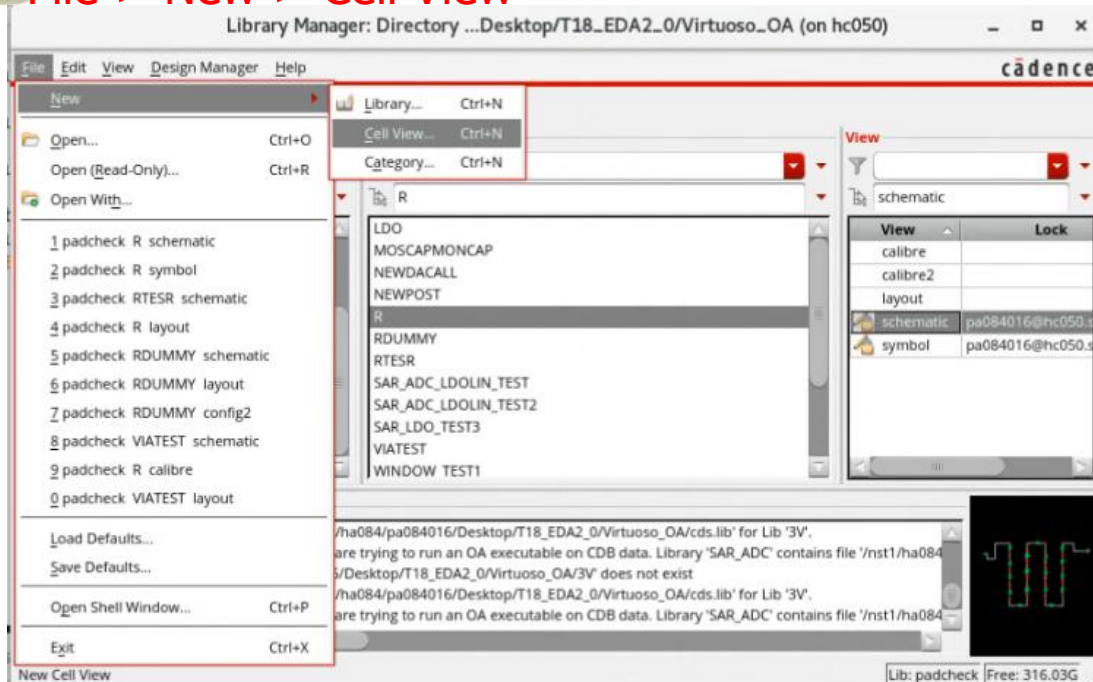


save and close when done

Create Testbench (1/3)

Return to the **Library Manager**

File > New > Cell View



Press i (Cre)



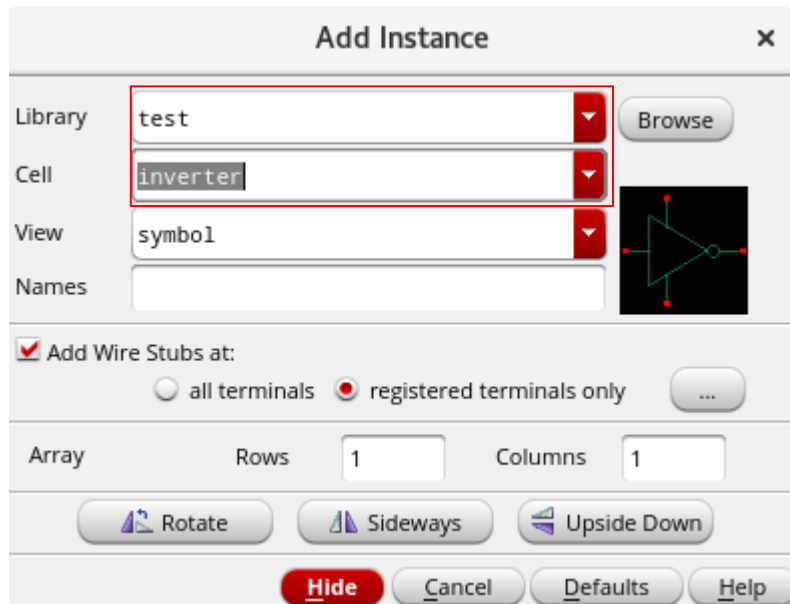
Source type : dc

GND

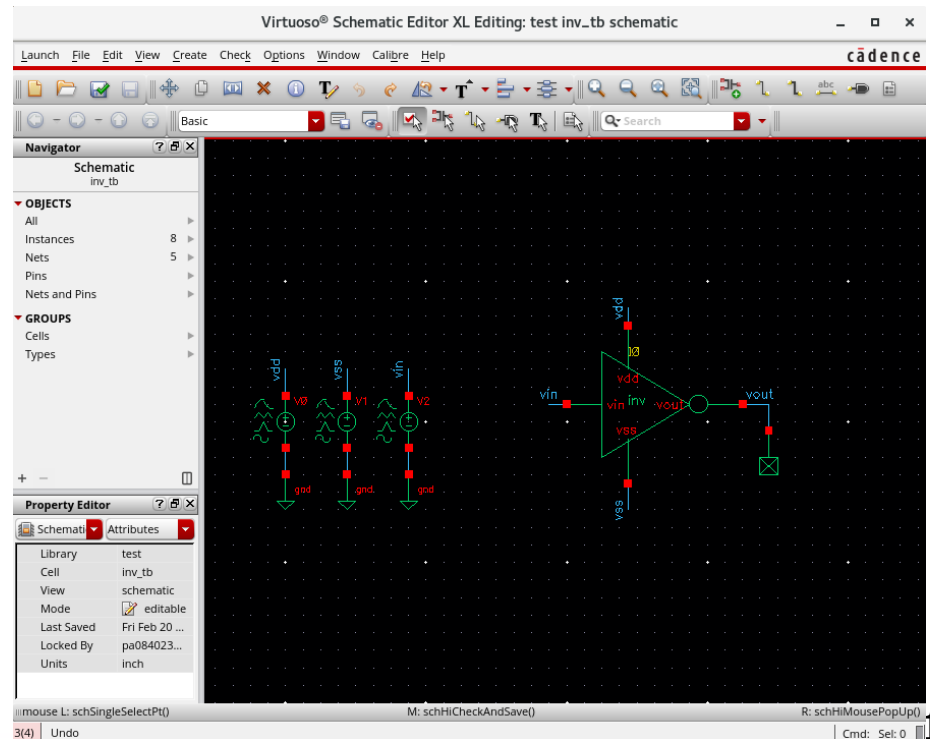
VDD15

Create Testbench (3/3)

press “i”

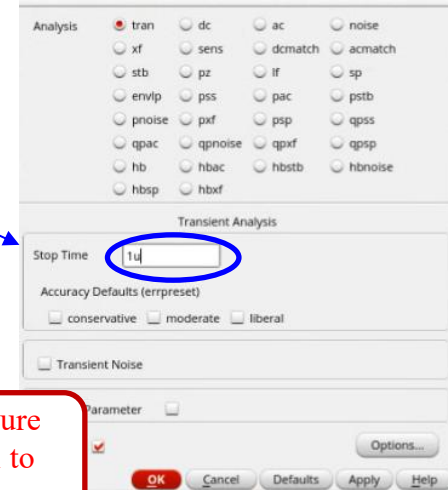
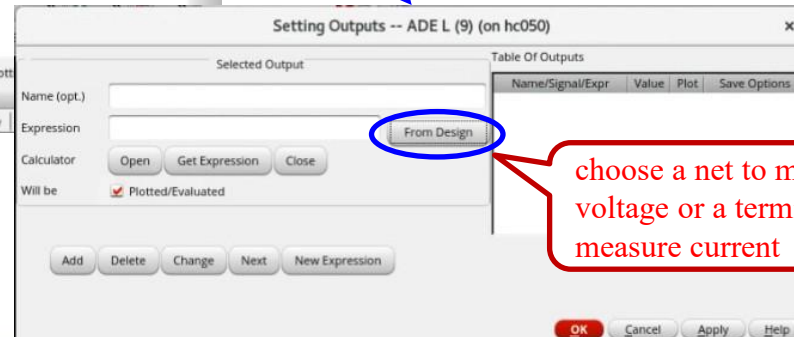
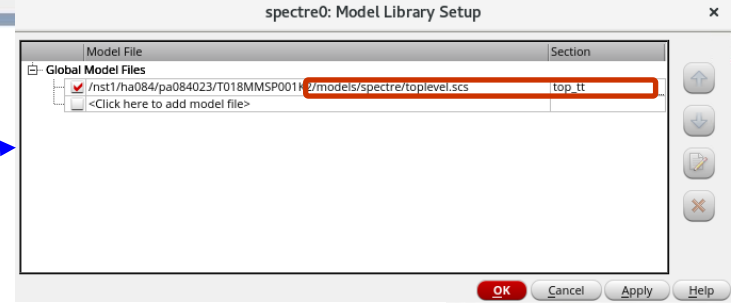
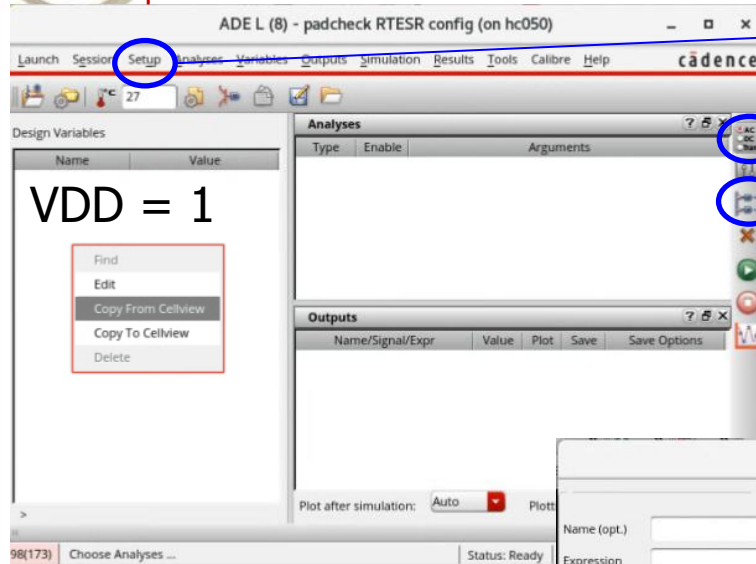


Launch > ADE L



Pre-Layout Simulation (Pre-Sim)

Setup > Model Libraries



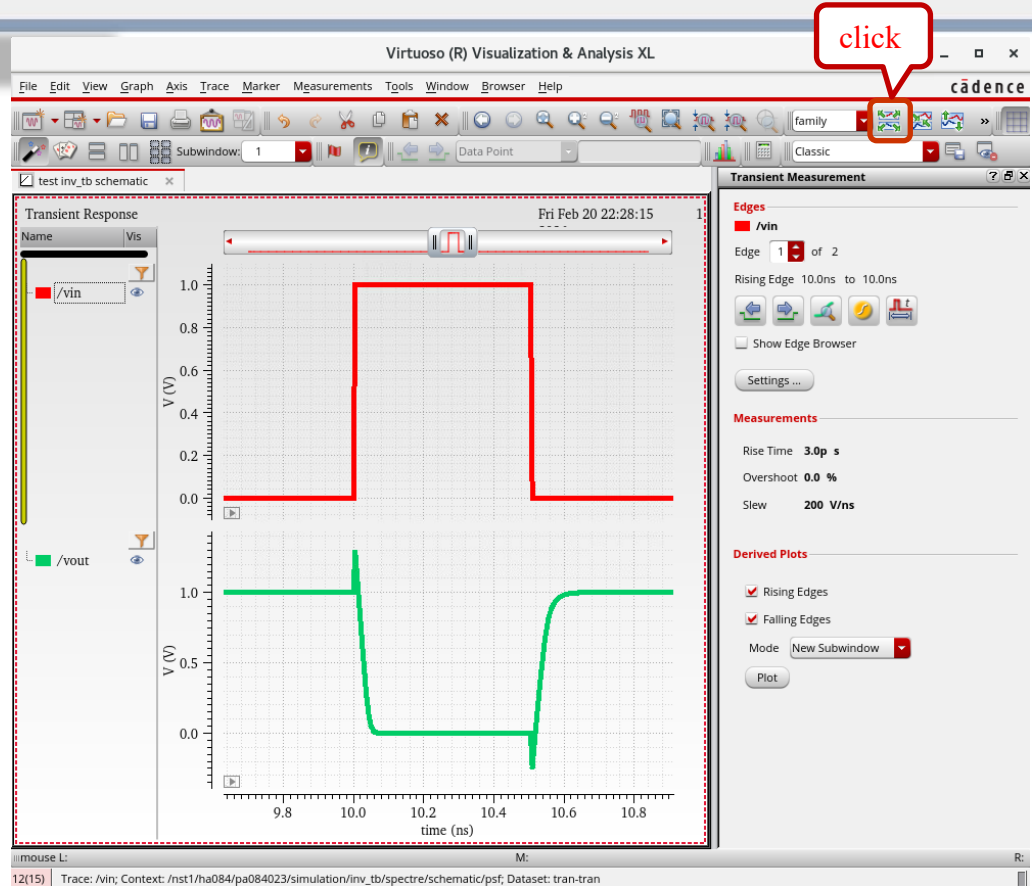
Pre-Sim Result

to make the plot look clearer

- Split All Strips
- File > Window Properties > set background to white > OK
- Ctrl+Shift+A > right click on a plot > Style: Solid & Width: Thick

to check transient measurement

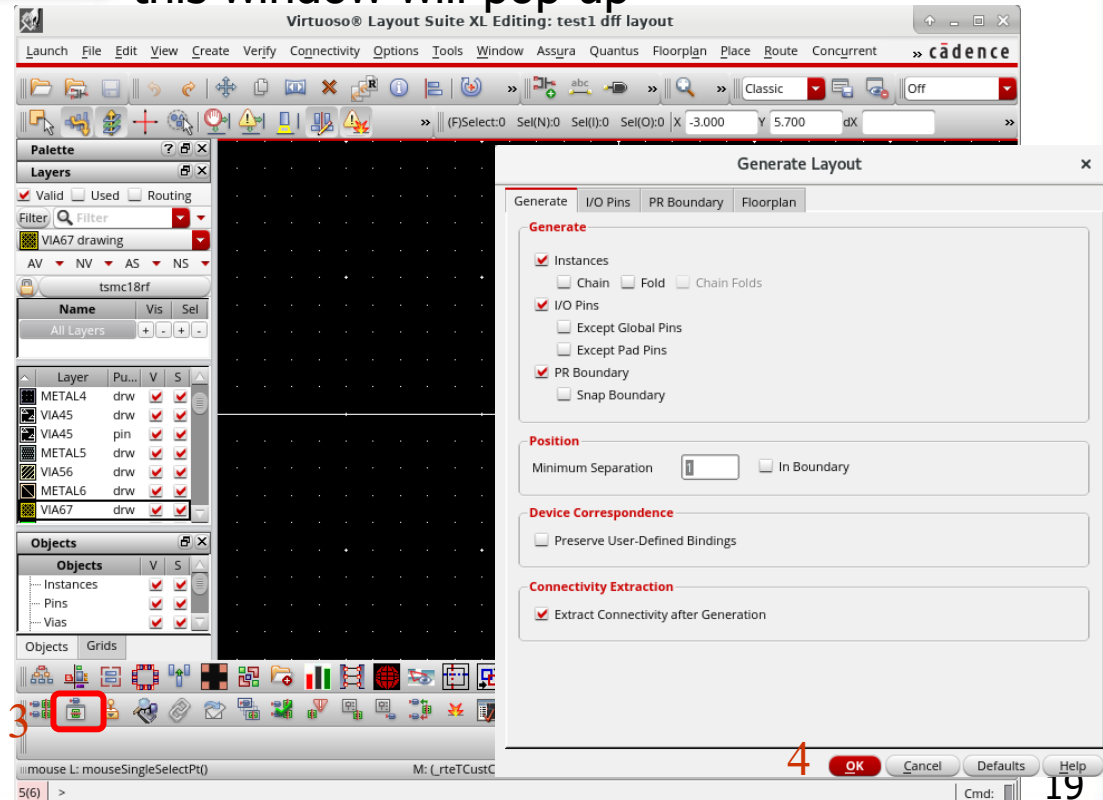
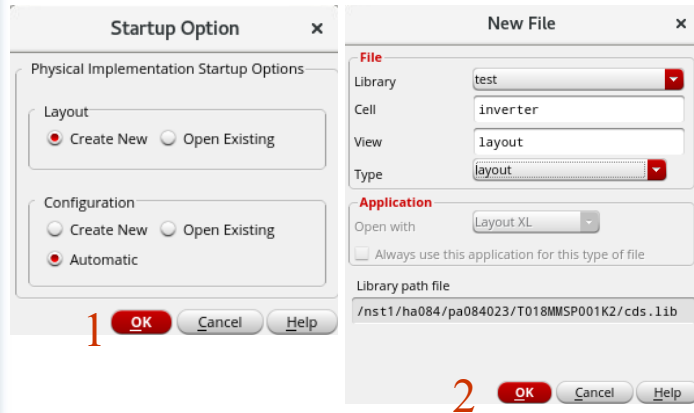
- Measurements > Transient Measurement



Virtuoso Layout Editor (a)

in Virtuoso Schematic Editor:
Launch > Layout XL

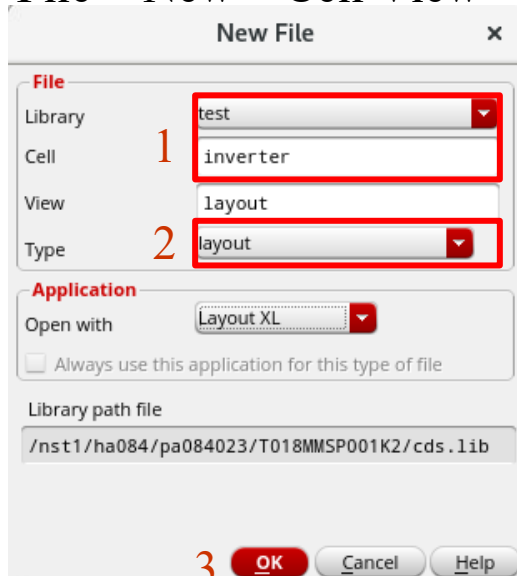
this window will pop up



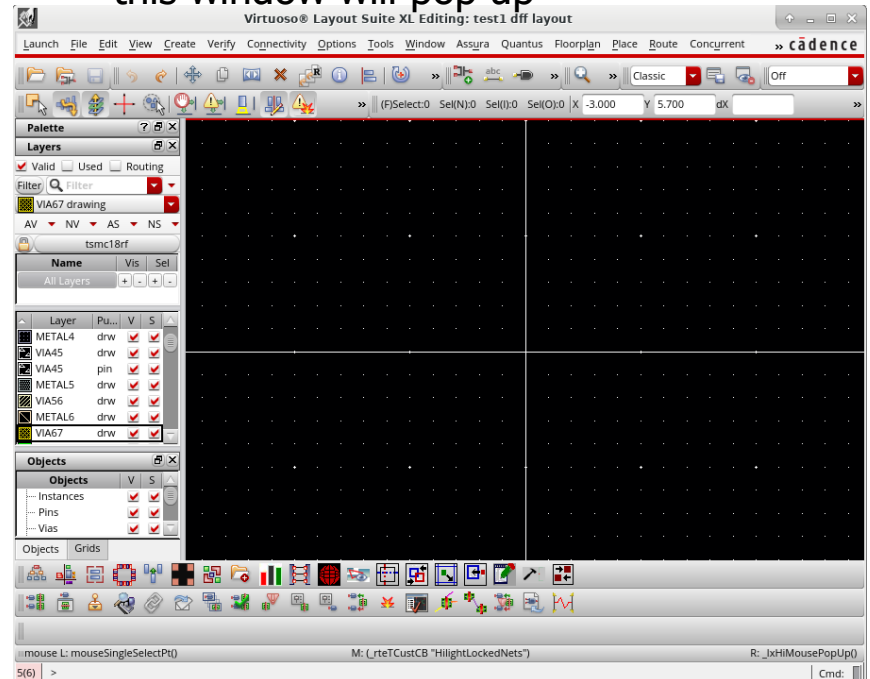
Virtuoso Layout Editor (b)

alternatively, you can draw the layout manually

1. return to the Library Manager
2. File > New > Cell View

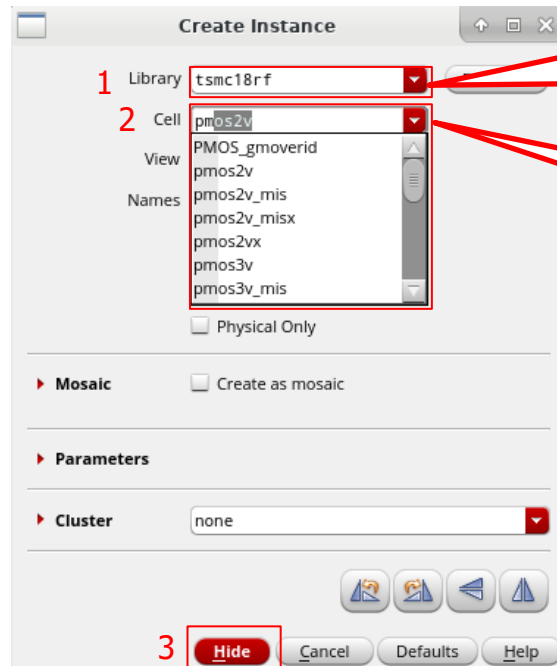


this window will pop up



Create Transistor (1/5)

- Press “I”



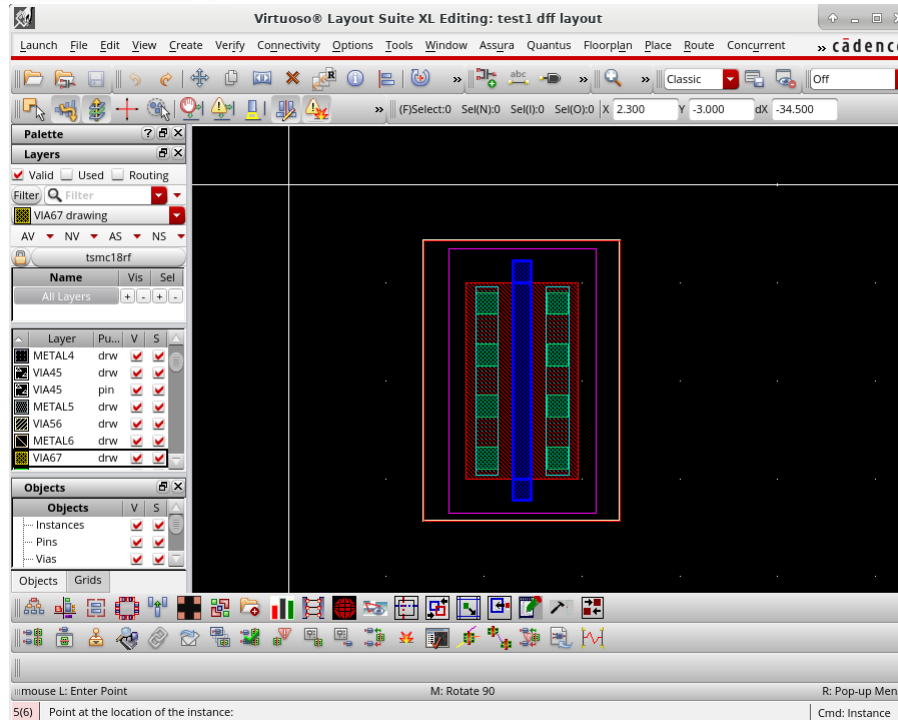
Select
“tsmc18rf”

Select “pmos2v” for
PMOS and “nmos2v” for
NMOS



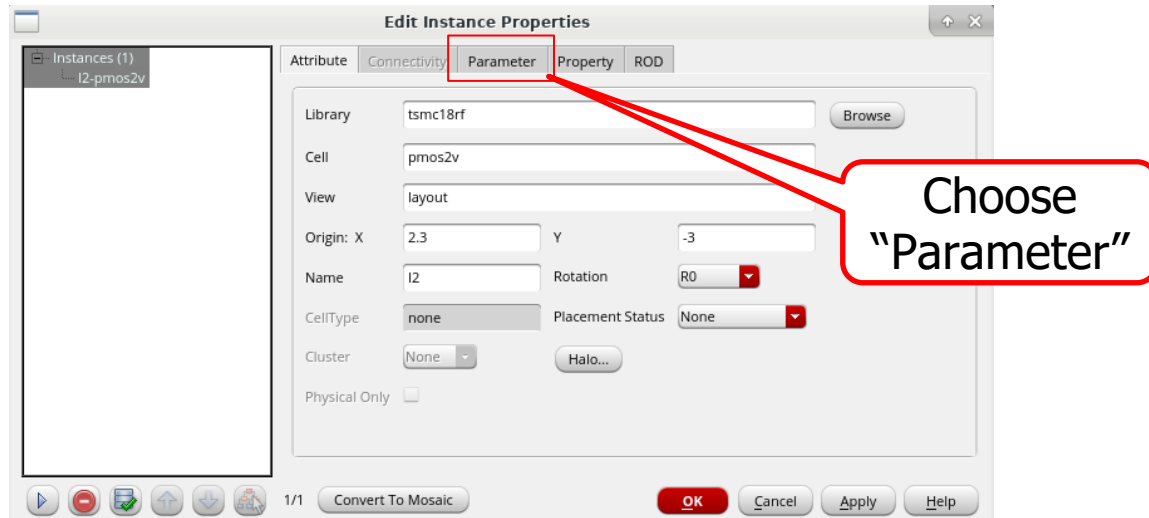
Create Transistor (2/5)

- Click anywhere to put the device

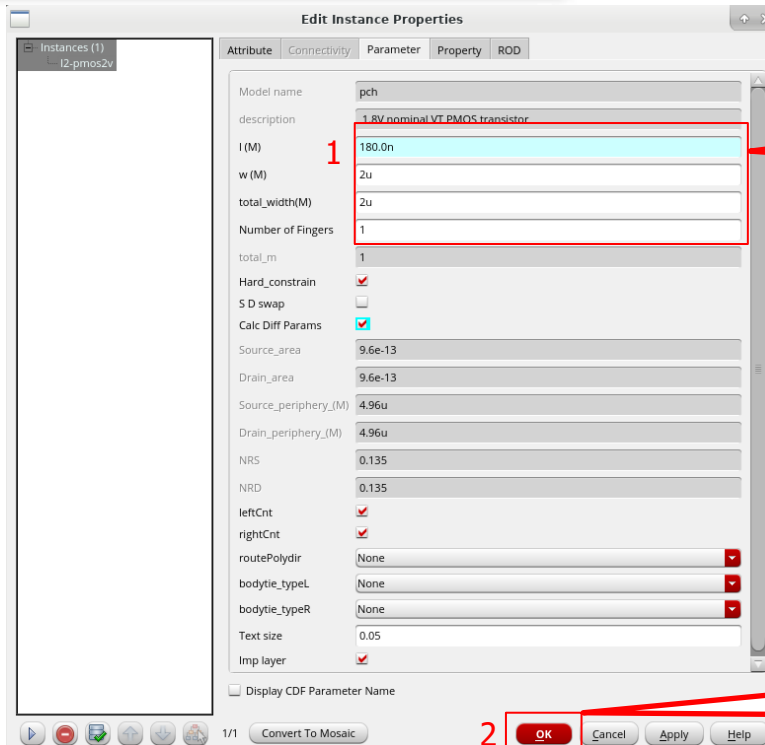


Create Transistor (3/5)

- Click on the device, then press “Q”



Create Transistor (4/5)

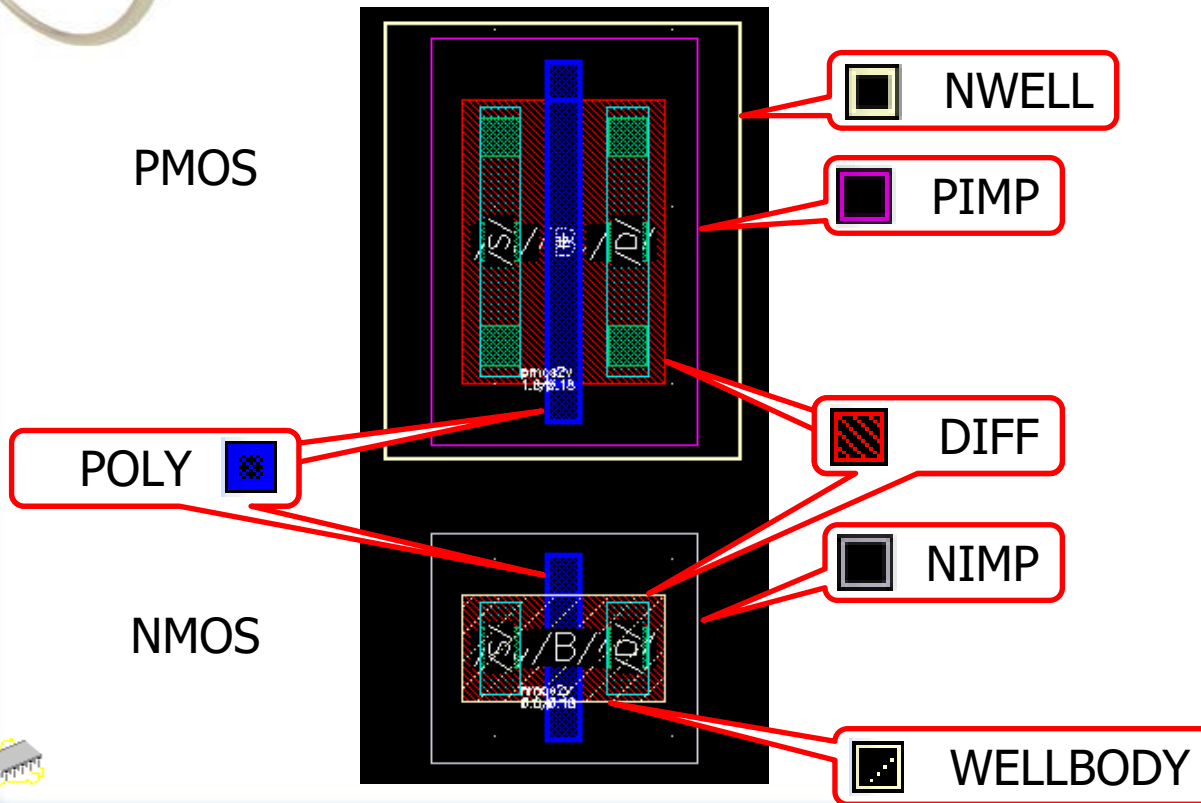


Edit the width, length, and number of fingers of your device here

Click "OK" to finish



Create Transistor (5/5)





Create Guard Ring (2/2)

You can create a guard ring by wrapping an area or using the interactive feature

you can also customize the Guard Ring setting manually

Creation Method

☐ Wrap

☒ Interactive

Shape

☐ Rectangle

☐ Polygon Ring

☐ Polygon Fill

☒ Path

Choose Device Automatically

Device: NGR

Contact Rows: 1

Path Width: 0.42

Net Name:

Wrap Settings: Wrap Type: placeAtMinDistance

Reference Layer Settings

Implant Layers Settings

Use	LPP	Enclosure Top/Right	Enclosure Bottom/Left	End of Line	Pin
☒	NIMP drawing	0.02	0.02	0.02	☐
☒	NWELL drawing	0.43	0.43	0.43	☐

Outer Rings: 0

More options

Main Layer: Diffusion

Contact Spacing: 0.25

Contact Dimension: 0.22

☐ Match Contact Enclosures

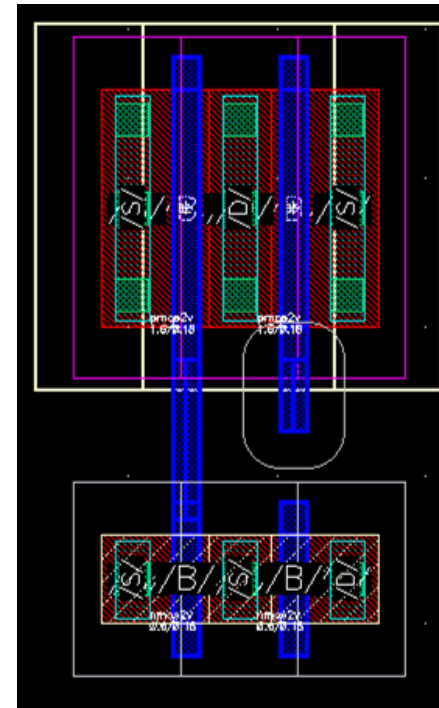
	Inside/Top/Right	Outside/Bottom/Left	End of Line
Diffusion Enclosure	0.1	0.1	0.1
Metal Enclosure	0.005	0.005	0.005

Hide Cancel Defaults Help

N-type Guard Ring (NGR) is for VDD, P-type Guard Ring (PGR) is for VSS

Create the connections

- Press “P” to create a path
- Click on POLY or metal layers to extend the path
- Connect all the terminals (S,G,D) according to the stick diagram
- Alternatively, press “R” to create a rectangle which also works as a connection



Create via (1/2)

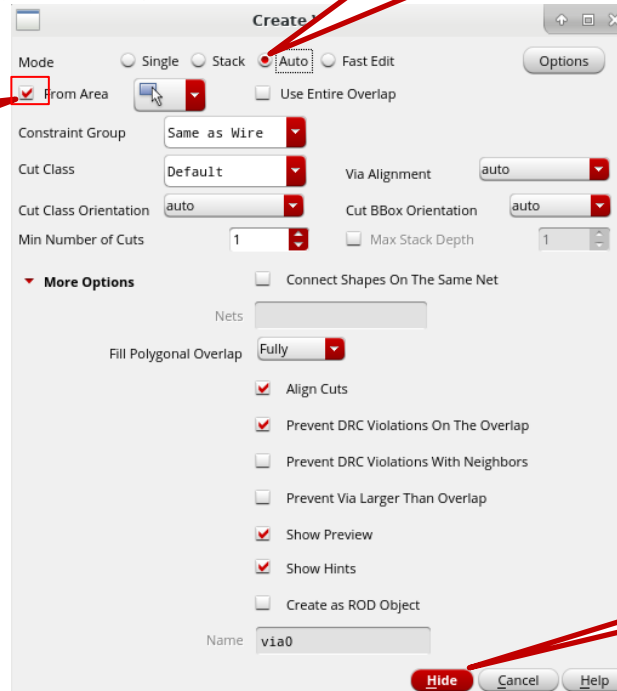
- Press “O” to create via

1

Choose “Auto”

2

Check “From Area”



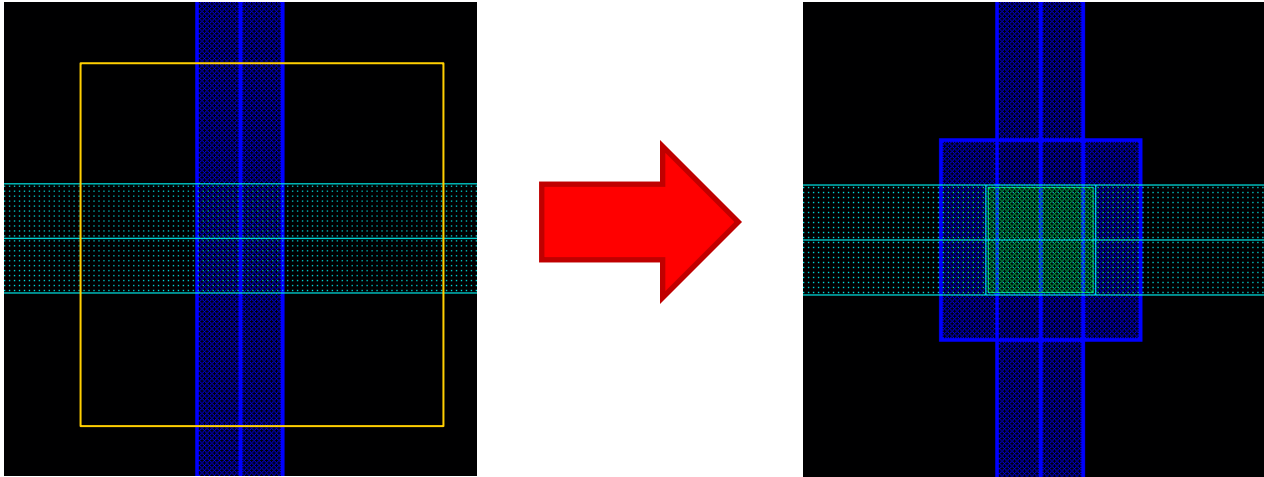
Click “Hide”

3



Create via (2/2)

- Select the overlap area between two layers



Create Pin

Press “L”

1

2

3

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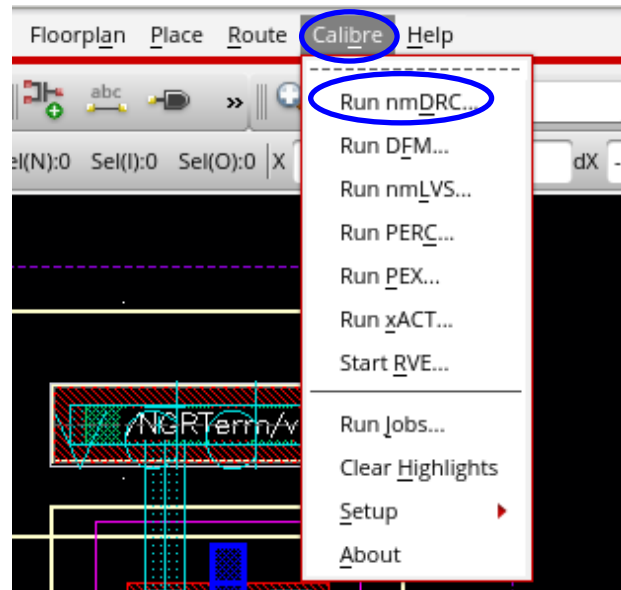
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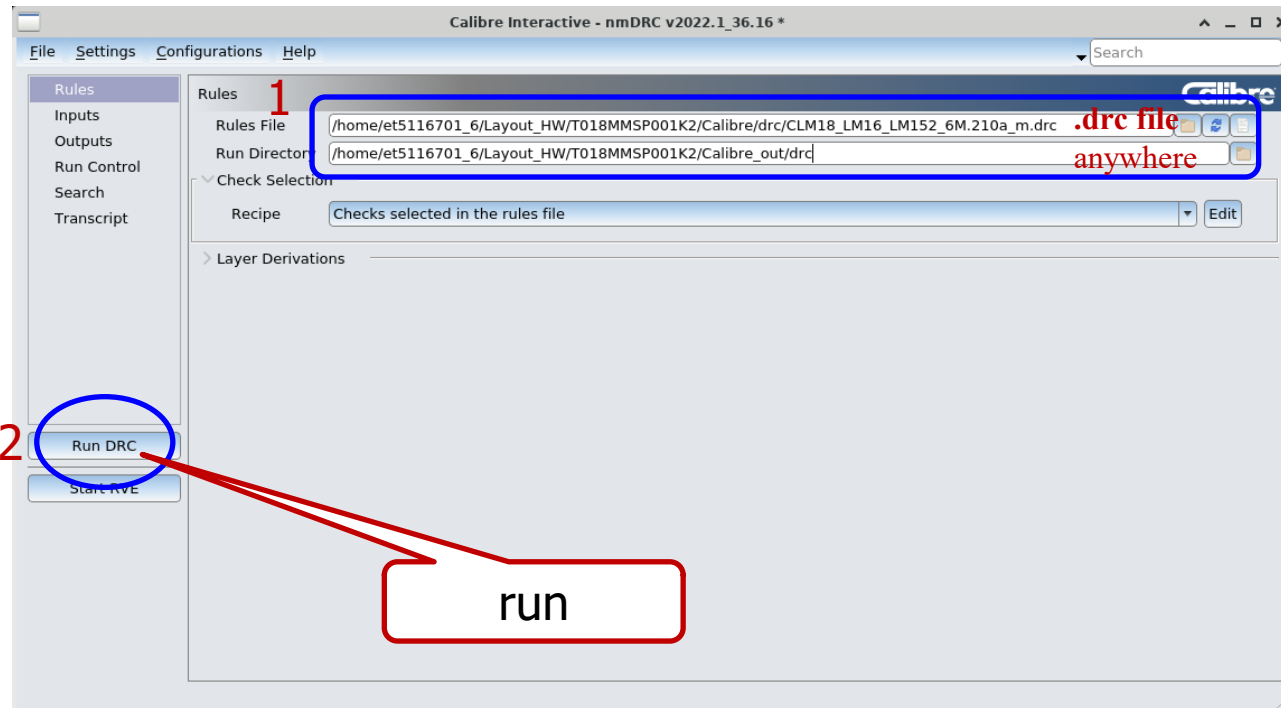
1488

Design Rule Check (DRC)

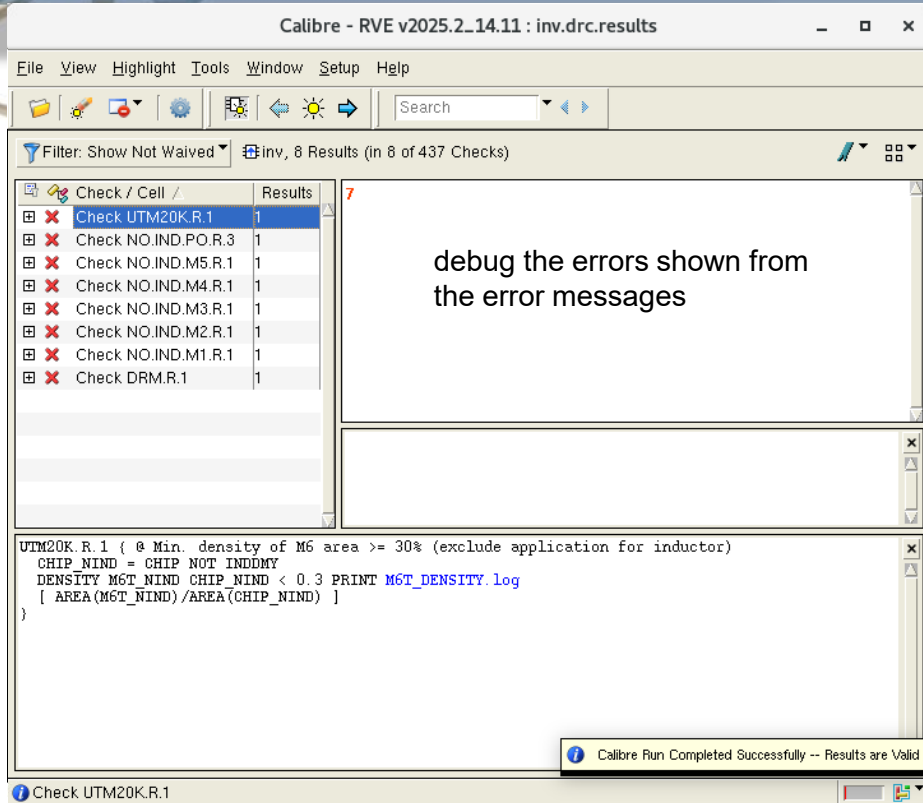


Design Rule Check (DRC)

Choose rules files & directory > click Run DRC



Design Rule Check (DRC)



There are error messages you can ignore which are mentioned in the PDF "chipservice_T18.pdf"

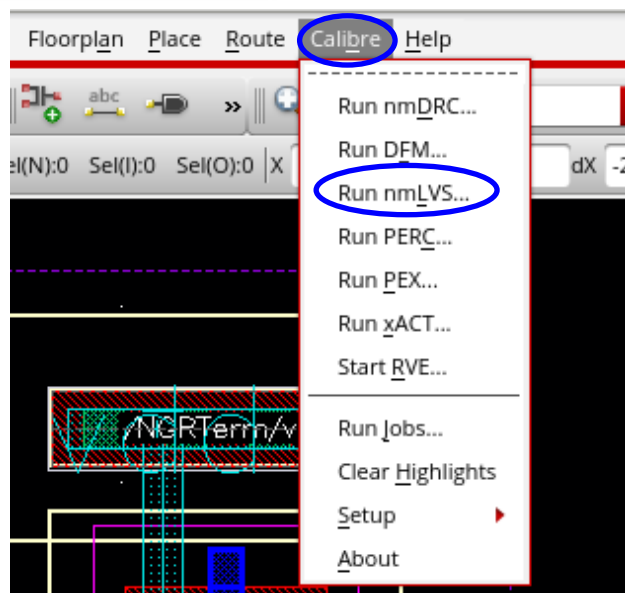


chipservice_T18





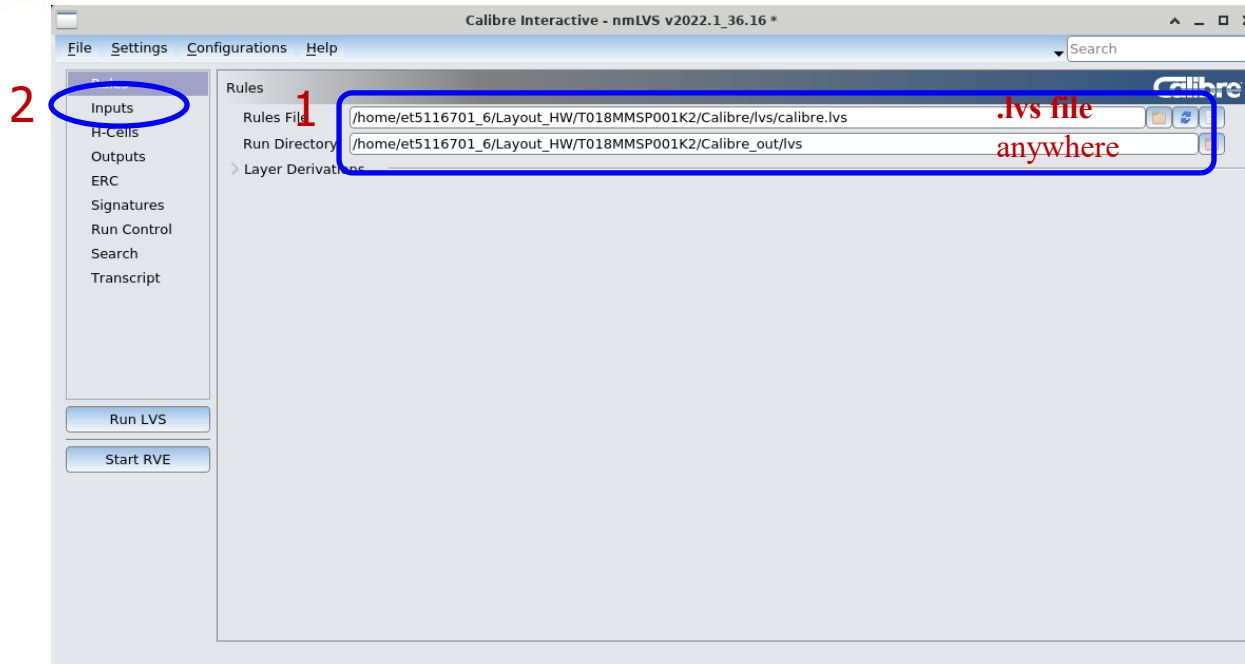
Layout Versus Schematic (LVS)



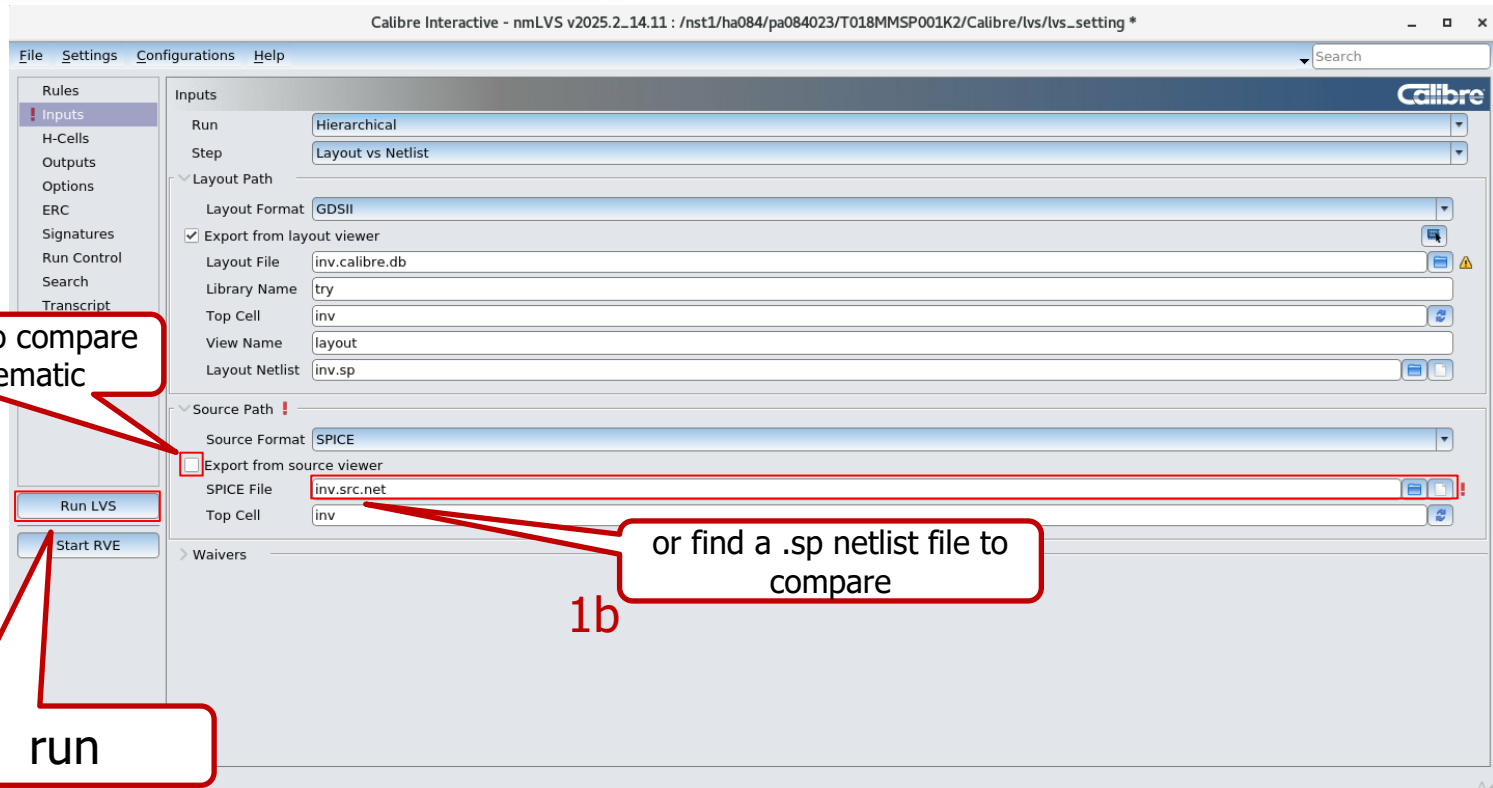


Layout Versus Schematic (LVS)

Choose rules files & directory > click Run LVS



Layout Versus Schematic (LVS)





Layout Versus Schematic (LVS)

A smiley face means it's passed :)

otherwise, happy debugging!

Calibre - RVE v2025.2_14.11 : svdb inv

File View Highlight Tools Window Setup Help

Navigator Info

Results

✓ Extraction Results

😊 Comparison Results

ERC

✓ ERC Results

ERC Summary

Reports

Extraction Report

LVS Report

Rules

Rules File

View

Info

Finder

Schematics

Setup

Options

Comparison Results

Layout Cell / Type	Source Cell	Nets	Instances	Ports
inv	inv	4L, 4S	1L, 1S	4L, 4S

Cell inv Summary (Clean)

CELL COMPARISON RESULTS (TOP LEVEL)

#

CORRECT #

#####

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⌘
⌘
⌘
⌘

LAYOUT CELL NAME: inv
SOURCE CELL NAME: inv

INITIAL NUMBERS OF OBJECTS

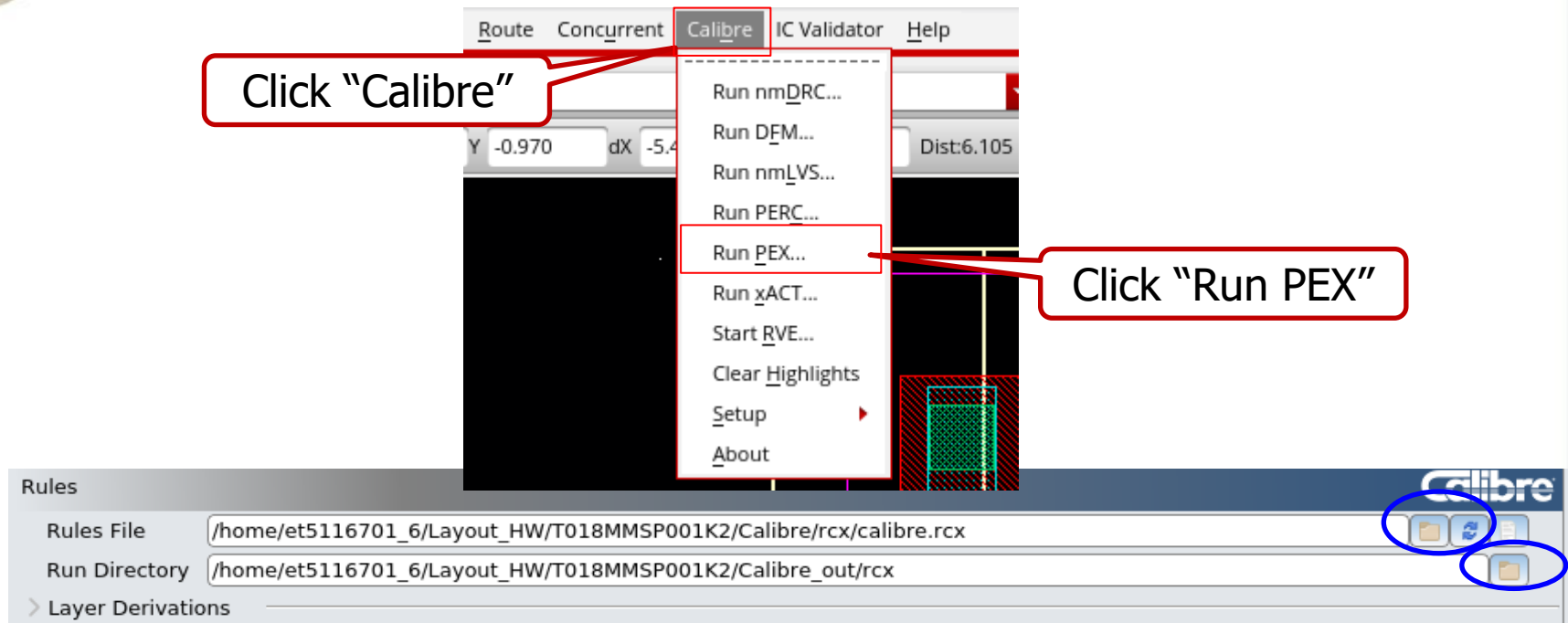
Layout

Source

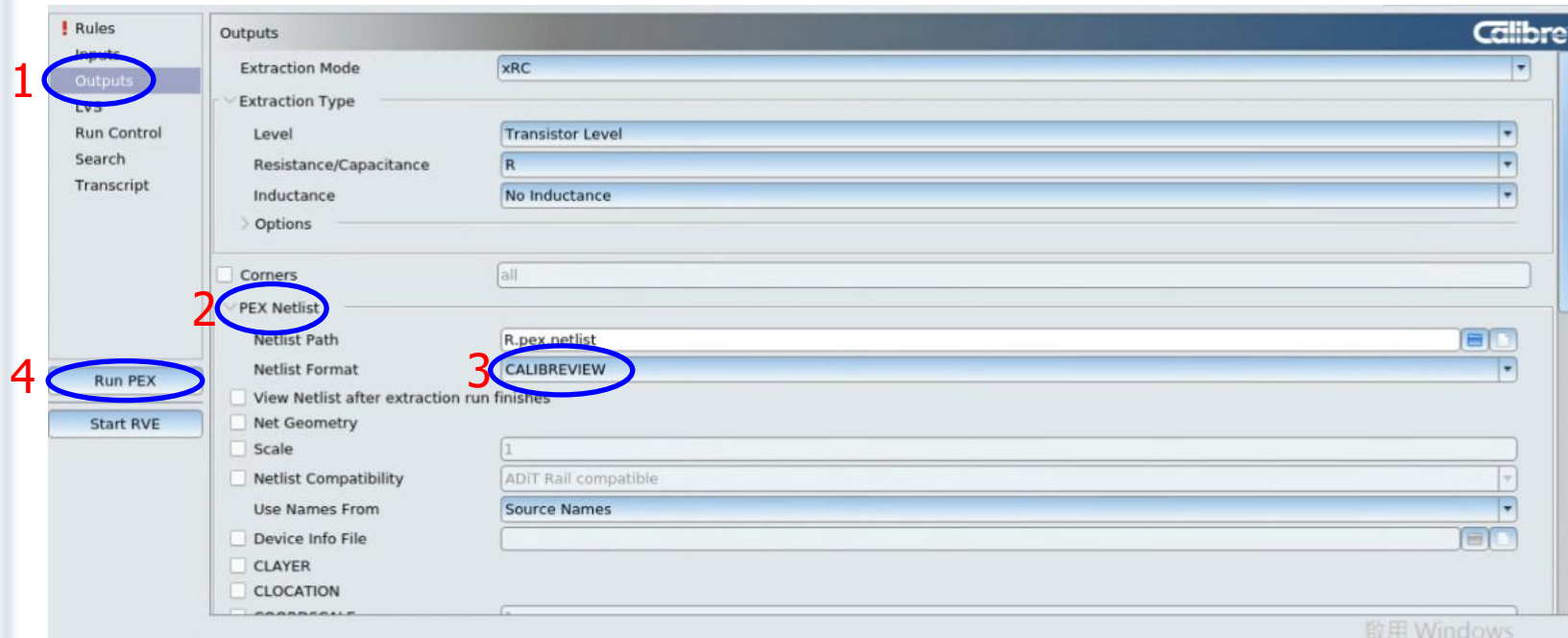
Component Type



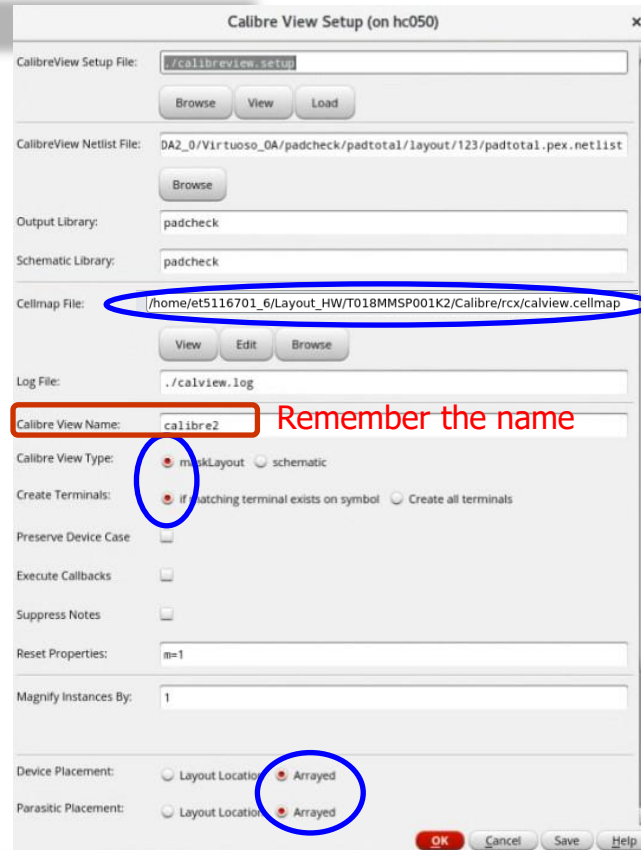
Parasitic Extraction (PEX)



Parasitic Extraction (PEX)



Post-Sim (1/4)



Calibre View Setup (on hc050)

CalibreView Setup File:
Browse View Load

CalibreView Netlist File: DA2_0/Virtuoso_OA/padcheck/padtotal/layout/123/padtotal.pex.netlist
Browse

Output Library: padcheck

Schematic Library: padcheck

Cellmap File:
View Edit Browse

Log File: ./calview.log

Calibre View Name: Remember the name

Calibre View Type: ☒ maskLayout ☐ schematic

Create Terminals: ☒ if matching terminal exists on symbol ☐ Create all terminals

Preserve Device Case ☐

Execute Callbacks ☐

Suppress Notes ☐

Reset Properties: m=1

Magnify Instances By: 1

Device Placement: ☐ Layout Location ☒ Arrayed

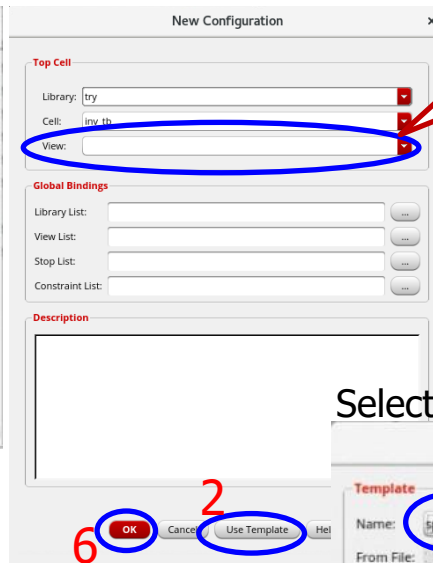
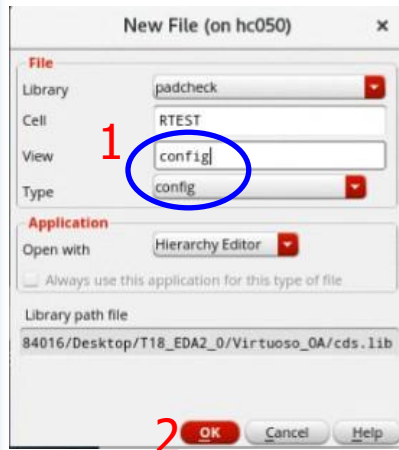
Parasitic Placement: ☐ Layout Location ☒ Arrayed

OK Cancel Save Help

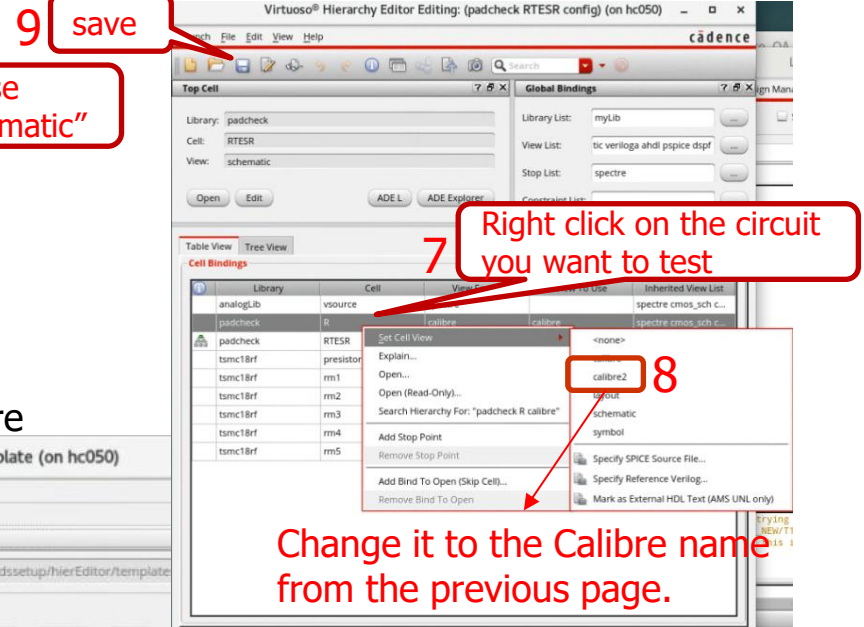
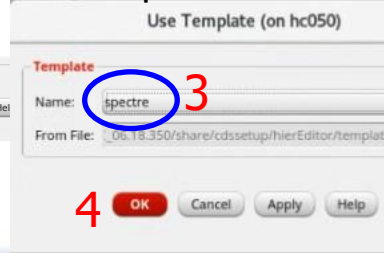
find this file

Post-Sim (2/4)

Return to **CIW** to create **config view**, Click File > New > Cellview
Select Library > Cell (**testbench**) > Type: **config** > View: **config**, click OK.

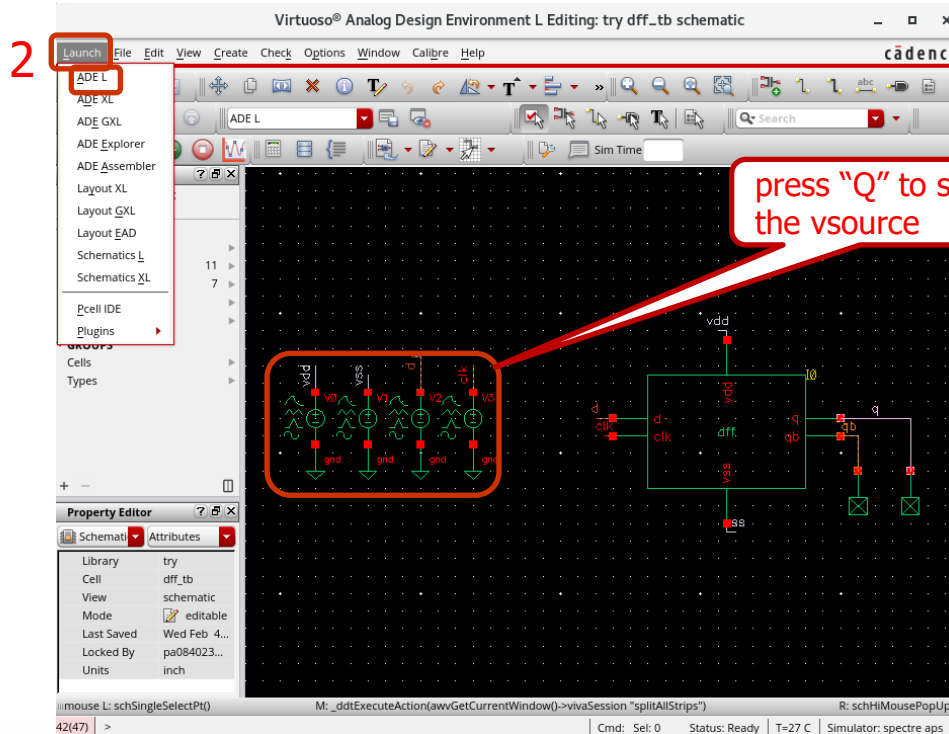


Select spectre



Post-Sim (3/4)

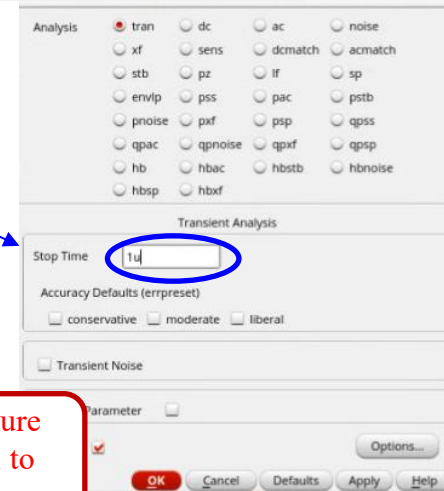
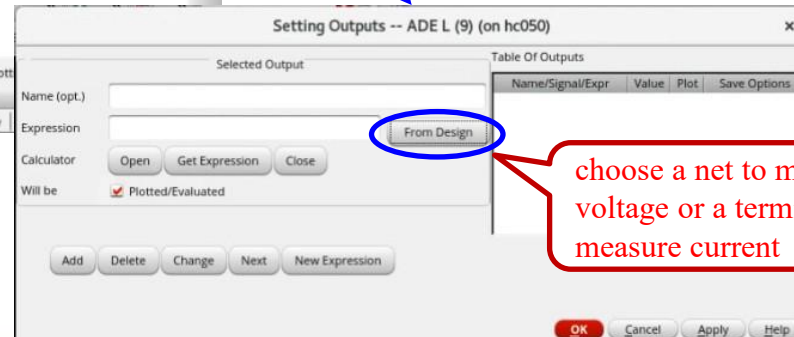
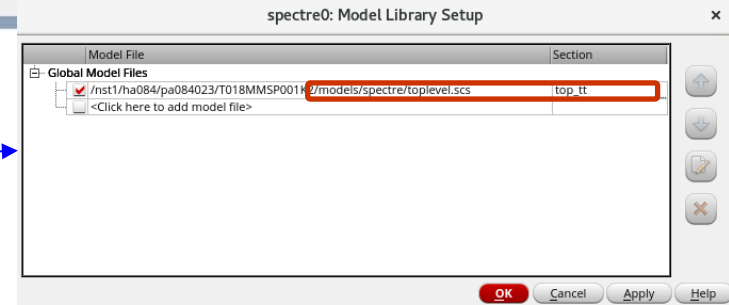
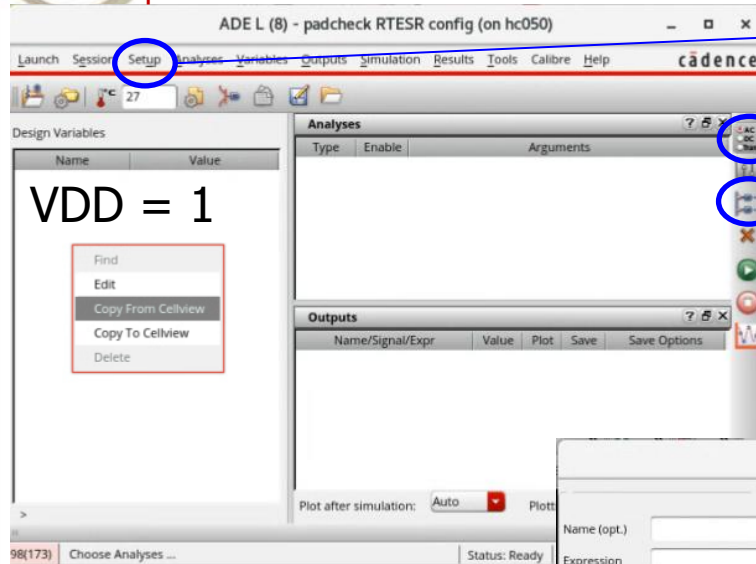
Open the config file > launch > ADE L



- 1 vdd
 - DC Voltage: VDD
 - Source type: dc
- vss
 - DC Voltage: 0
 - Source type: dc
- d
 - Source type: pulse
 - Zero value: 0
 - One value: VDD
 - Period of waveform: 100n
- clk
 - Source type: pulse
 - Delay time: 20n
 - Zero value: 0
 - One value: VDD
 - Period of waveform: 20n

Post-Sim (4/4)

Setup > Model Libraries



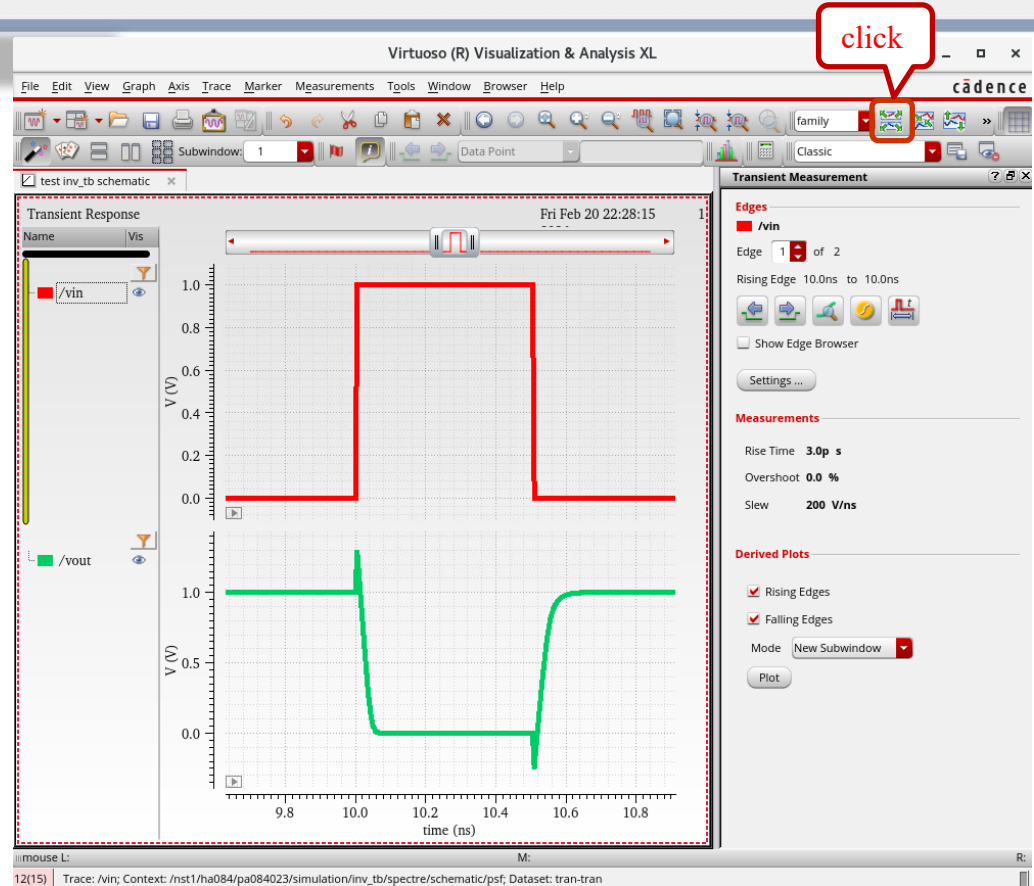
Post-Sim Result

to make the plot look clearer

- Split All Strips
- File > Window Properties > set background to white > OK
- Ctrl+Shift+A > right click on a plot > Style: Solid & Width: Thick

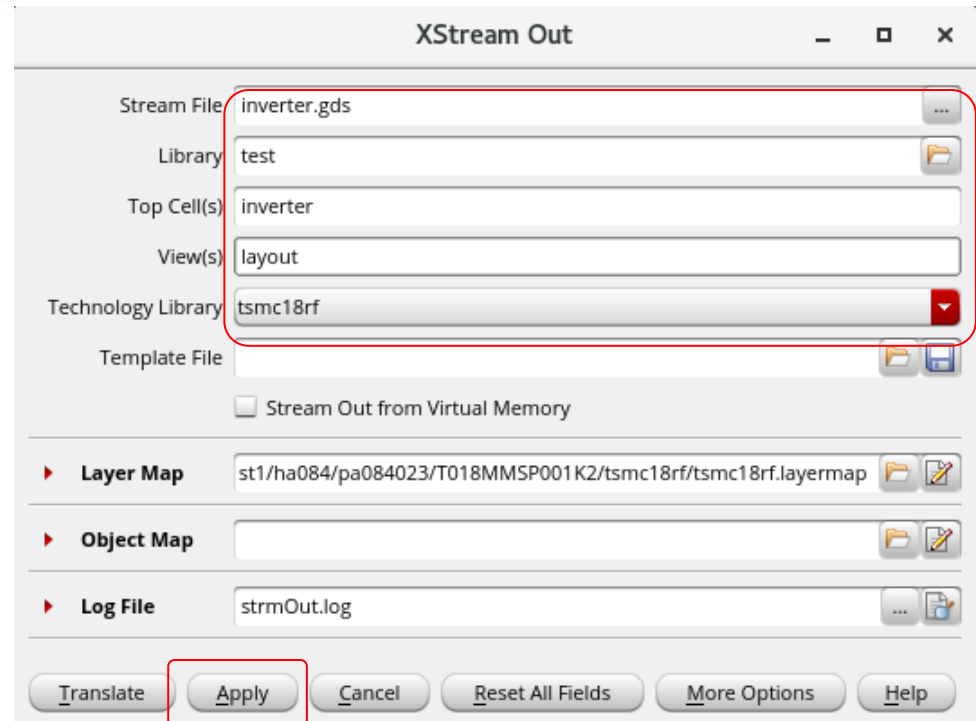
to check transient measurement

- Measurements > Transient Measurement



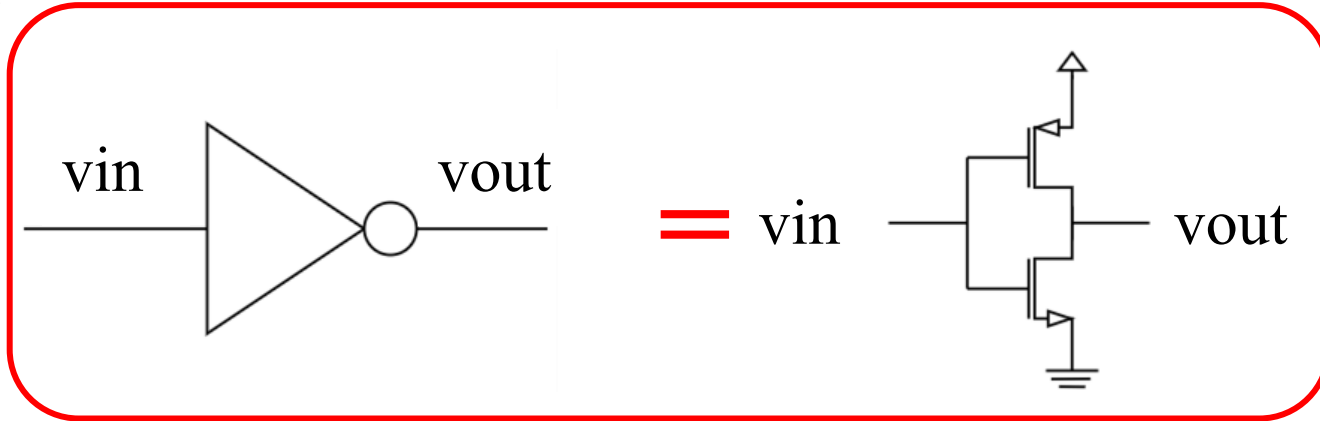
Import gds file

in the CIW, choose
File > Export > Stream...



Homework: Inverter

- Top Cell Name: inverter



Pin names: vin, vout, VDD, VSS





Homework: Size & Performance

■ Size

Device	Width (um)	Length (um)	Multiplier
PMOS	1.6	0.18	1
NMOS	0.6	0.18	1

■ Measure

Performance	Better
Power consumption	lower
layout area	lower
rise/fall time	symmetric, lower





Scoring

- The score criterion is based on the design of the completeness of each student. The layout should be verified correctly by DRC/LVS tools. If Not, the total score is 0 point.
- The score will be calculated by the formula below:
- Use only Poly & Metal 1 for connections.
- $\text{Score} = (\text{layout area}) * 0.5 + (\text{hand in time}) * 0.2 + (\text{post sim result}) * 0.2 + (\text{report}) * 0.1$
- The earlier you turn in your homework, the higher your score will be.





Report

- Please follow the report format provided. Only PDF files are accepted.
- The report must be written in **English** and should be **no longer than two pages**.
- The report must include the **layout diagram, area information, and DRC and LVS reports**.
- Please submit your report via **Moodle**.
- The submission time recorded by Moodle will be used to determine whether the report is submitted on time.
- Please submit the **report** file within two weeks. (Deadline: October 2, 11:59 PM)
- The report files should be compressed into a single file and named using the following format: Lab1_“StudentID”_“StudentName” Example: **Lab1_m11402241_劉丞皓.pdf**

